

DIAGNOSIS AND TREATMENT METHODS OF BEE DISEASES

Editor **Münire TURHAN | Ebubekir İZOL**

Bee diseases are seen as an important problem for beekeepers and scientists working in this field. There are many bee diseases, and these diseases cause mass deaths of bees. In order to completely combat these diseases, a lot of research is being carried out in this field. For this reason, in this book titled Diagnosis and Treatment Methods of Bee Diseases, bee diseases are explained by experts in the field, and current methods and solution techniques are included. This book aims to make bee diseases and possible treatment methods known. We hope that this book will be an important resource for beekeepers and scientists working in this field, especially for those interested in the diagnosis and treatment methods of bee diseases.

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 **NOBEL**

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Diagnosis and Treatment Methods of Bee Diseases

Editors

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PREFACE

Bee diseases are seen as an important problem for beekeepers and scientists working in this field. There are many bee diseases, and these diseases cause mass deaths of bees. In order to completely combat these diseases, a lot of research is being carried out in this field. For this reason, in this book titled *Diagnosis and Treatment Methods of Bee Diseases*, bee diseases are explained by experts in the field, and current methods and solution techniques are included. This book aims to make bee diseases and possible treatment methods known. We hope that this book will be an important resource for beekeepers and scientists working in this field, especially for those interested in the diagnosis and treatment methods of bee diseases.

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BIOLOGICAL RESOURCES AND CHEMICAL DRUGS USED AGAINST BEE DISEASES

Ebubekir İZOL

Bees are arthropod insects belonging to the Apoidea family that feed mainly by collecting nectar and pollen from flowers. Bees usually live in communities and are known for their social organization. Bees are important pollinators in ecosystems and play an important role in maintaining the balance of nature and increasing productivity in agricultural ecosystems. Declines in bee populations have been observed in recent years. Among the many reasons for these declines, bee diseases have an important place. However, various diseases threaten bee populations and negatively affect the beekeeping industry. This leads to a decline in bee products and the destabilization of natural vegetation and ecosystems. Bee colonies are being lost due to diseases, and this is becoming a major problem. Thousands of bees perish every year in our country and around the world due to diseases. This situation causes many different negatives besides economic loss. For this reason, scientific studies are carried out in many different fields to prevent bee deaths worldwide.

In this chapter, bee diseases are defined, and an overview of biological products and chemical drugs used to prevent bee diseases is given. Many different

methods and materials are used to prevent bee diseases. Some of these are natural products of plant origin, while others are synthetic products of chemical origin.

Some Significant Bee Species

1. Honey Bees (*Apis* spp.):

One of the best-known bee species and the most widely used species is *Apis mellifera*. They are important to humans because of their social organization and honey production. They usually live in hives and collect mostly nectar and pollen (1).

2. Bumble Bees (*Bumble* spp.):

They are soil-dwelling bees that collect nectar from flowers and carry pollen. They have social organization but not as complex as honey bees. They play an important role in pollinating mostly wild plants (2).

3. Lonely Bees (Spider and Fly Bees):

They are mostly solitary and have no social organization. They make their own nests by collecting nectar and pollen from flowers. They play an important role in plant reproduction by carrying pollen (3).

4. Unsocial Bees (*Trigona* and *Melipona* spp.):

Bee species with social organization but not economically important in terms of honey production. They are usually found in tropical regions and build their nests in tree hollows. They feed their nests by collecting nectar and pollen (4).

Each bee species is unique in its ecological roles and behaviors and plays an important role in ecosystems. By studying the biology and ecology of these bee species, scientists and researchers are trying to conserve wildlife and improve the beekeeping industry.

Bee Diseases

Bee diseases occur under the influence of various pathogens (viruses, bacteria, fungi) that cause physiological and behavioral disorders of bees. These diseases can spread in bee colonies and are usually transmitted between infected individuals through contact or food. The effects of bee diseases can have a

negative impact on colony strength, honey production and overall health. Colonies exposed to diseases can be less productive and even die out. This can negatively affect pollination services in the agricultural industry and cause economic losses. Various methods are available for the control and treatment of bee diseases. These include chemical drugs, biological control agents and cultural practices. However, more research is needed on the efficacy and side effects of treatment options (5,6).

Bees are susceptible to a number of diseases caused by various pathogens. These diseases include:

1. Varroa Destructor:

Varroa mite is a parasite that causes serious damage to bee colonies. By living in the bodies of bees, it both sucks the blood of bees and can cause the spread of viral diseases. Varroa mites cause disease and death in bees through 2 main mechanisms. The first mechanism is the loss of fat body stores leading to the death of infected brood. The second mechanism is the transmission of viral diseases. Varroa mites have been identified as vectors of 8 viruses (7).

Treatment Methods: The most commonly used methods of controlling Varroa include chemical pesticides, the use of organic acids, thermal control methods and mechanical control (e.g. boring bees by flying them). It is also important to breed Varroa-resistant hybrids for bees and apply integrated bee management techniques.

2. Nosema:

Nosema is a single-celled parasitic protozoan (*Nosema apis* or *Nosema ceranae*) that lives in the intestines of bees. Nosema infection can affect the digestive system of bees, causing weakening and death in colonies. Both strains of Nosema are strongly linked to stress and are best prevented by maintaining strong, healthy colonies. The diagnosis of Nosema is confirmed by microscopic examination of the bee's hindgut contents and identification of spores typical of Nosema spp.

Treatment Methods: Antiparasitic drugs, gut health supplements such as probiotics and prebiotics, hygienic practices and measures to reduce bee stress can be used to combat Nosema. Pollen supplements and sugar syrups may also be recommended to improve the nutrition of colonies and

strengthen overall health (8). Other treatment options for *Nosema* include feeding plant phytochemicals and probiotics to strengthen the honey bee gut microbiome. Drugs used in other countries to treat *N. ceranae* in honey bees, such as metronidazole or tinidazole, are prohibited for use in the US.

In a study, the effects of natural compounds (thymol, vetiver essential oil, lysozyme, resveratrol) on honey bees infected with *nosema* disease were evaluated. For this purpose, bees were dosed with *nosema* spores separately and fed with candies prepared with the screened substances. As a result, it was found that the infection rates of bees fed with thymol and resveratrol sugars were significantly lower and bees fed with sugars prepared with resveratrol lived significantly longer. This study suggests that thymol and resveratrol may be useful in alternative strategies for the control of *nosema* disease (9).

Many compounds have been tested for *Nosema* control, but the only effective product is the antibiotic fumagillin, which inhibits the development of *N. apis* in the honey bee (10-12). Since the use of antibiotics for the treatment of diseased hives is prohibited in most European countries, control of the disease is mainly achieved by applying preventive measures such as placing hives in non-humid areas, transferring bees to non-contaminated equipment and fumigation of combs with acetic acid to kill *N. apis* spores. Even in cases where treatment with fumagillin is possible, there is the problem of disease recurrence, since according to some authors only the vegetative forms of the parasite are killed (13-15), while Liu (1973) suggests that fumagillin can alter the ultrastructure of the membrane in young spores. Although there are no reports of fumagillin residues following treatment of *Nosema* disease, this antibiotic has been shown to be very stable in honey (16).

3. American Foulbrood:

American foulbrood is an infectious disease caused by a bacterium called *Paenibacillus larvae*. This disease causes the death of bee larvae and can reduce the strength of the colony. American foulbrood can be detected in suspect cells (discolored, collapsed, perforated or partially closed cells) using the ropiness test, which consists of inserting a toothpick or tweezers and twisting the cell contents. The disease is confirmed if the cell contents twist out of the cell and return to the cell when the thread breaks.

Treatment Methods: Antibiotics (e.g. oxytetracycline) can be used to control American foulbrood. Also important are cultural practices such as removing infected larvae and disinfecting hives to improve the hygiene of the colony (17).

4. European Foulbrood:

European foulbrood is an infectious disease caused by the bacterium *Melissococcus plutonius*. This disease can cause the death of bee larvae and affect the overall health of colonies. European foulbrood is seasonal; it usually occurs in early spring before the start of the nectar flow, disappears during the nectar flow and in summer, and reappears in the fall. European foulbrood can usually be prevented by maintaining colonies that are large, healthy and not subject to stress (18).

5. Black Foulbrood:

Black foulbrood rot is an infectious disease caused by the fungus *Ascosphaera apis*. This disease can cause black colored rot in the digestive tract of bee larvae, leading to the death of the larvae (6).

6. Wax moths:

Large wax moths (*Galleria mellonella*) and small wax moths (*Achroia grisella*) do not directly harm adult bees or larvae but can cause damage, especially to comb, honey and hive woodwork. Wax moths are not usually a problem in strong honey bee colonies but can be easily detected by silky tunnels on the comb and evidence of black fecal droppings. Using naphthalene to prevent wax moth is wrong. Because naphthalene is a carcinogenic substance and can leave residues in honey (19,20).

7. Deformed Wing Virus:

This virus can cause deformation of bees' wings and can seriously affect their ability to fly (21).

8. Small Hive Beetles (*Aethina tumida*):

First observed in the USA in the mid-1990s, small hive beetles are small black insects that are usually seen on the inside of a colony hive box when the inner lid is opened. Small hive beetles lay eggs in honey bee colonies and the secretions (slime) of the hatched larvae contaminate the honey, causing it to ferment and spoiling the honey harvest. Once a hive is contaminated with

small hive beetle mucus and the honey has fermented, the frames inside the hive should be removed for thorough cleaning, as the bees cannot remove them. Most beekeepers place insect traps filled with oil inside the hive to catch and drown small hive beetles. Outside the hive, small hive beetle larvae can be prevented from becoming pulp by wetting the ground under the hive with ground treatments such as permethrin. Neither the oil used in insect traps nor the application of permethrin to the floor under the hive has been linked to harmful chemical residues in honey (22).

These bee diseases can cause serious damage to the health of bee colonies and negatively affect the beekeeping industry. Therefore, beekeepers and researchers are constantly working on the prevention, identification and treatment of these diseases. While insecticides can directly kill these vital insects, herbicides indirectly affect their survival and reproduction by reducing the diversity of their food sources. At the non-lethal level, neurotoxic insecticide molecules are known to affect the cognitive abilities of bees, impairing their performance and ultimately affecting the viability of colonies. In addition, common systemic insecticides appear to profoundly affect the health of both honey bees and wild bumblebees, leading to indirect side effects (23).

Many natural and chemical substances are used for the control of bee diseases. The effect of these substances can vary depending on the type of disease, the way they are used and environmental factors. These substances are used to maintain bee health and strengthen colonies.

Natural Substances Used in Bee Diseases

1. Propolis:

Definition: A bee product created by bees through the reaction of plant nectar, pollen, beeswax and bee enzymes. Propolis produced by bees has antimicrobial and antifungal properties. Beekeepers can prevent the spread of bacteria and fungi by coating their hives with propolis (24).

Mechanism of Action: Propolis can inhibit the growth of microorganisms and reduce the risk of infection thanks to the flavonoids and phenolic compounds it contains. Studies have shown that propolis is a very powerful antioxidant and thus plays a role in the healing of diseases (25).

2. Royal Jelly:

Description: Royal jelly is produced by bees for the nutrition of larvae. It can increase the resistance of bees against diseases with its nutrients and immune system strengthening properties (26).

Mechanism of Action: Royal jelly can support the immune system of bees through the proteins, vitamins and growth factors it contains and may have a protective effect against diseases (27).

3. Antibacterial Properties of Honey:

Definition: Some honeys have antimicrobial properties (28). Especially Manuka honey is a natural component used in the control of bee diseases.

Mechanism of Action: Honey can inhibit the growth of bacteria and fungi and prevent the spread of infections thanks to its methylglyoxal and other active components. Since the bioactive components in honey show antibacterial and antiviral properties, honey is effective against such diseases. (29).

Apart from these, many plant-derived fatty acids and organic acids are also used in bee diseases. Some of these are: Thymol, Hops beta acids, Amitraz, Coumaphos, Flumethrin, Fluvalinate (30). Recent studies in this field have shown that essential oils of plants such as thyme, clove, mint, lemongrass, cinnamon, grapefruit, rosemary, marigold are lethal against some mites, bacteria and fungi. In addition, some components isolated from these plants such as sanguinarine, thymoquinone, capsaicin, carvacrol, citral, eugenol, thymol have been reported to have these effects on organisms (31).

Chemical Substances Used in Bee Diseases

1. Oxytetracycline

Description: Used as an antibiotic, oxytetracycline is effective in the control of American foulbrood and other bacterial infections.

Mechanism of Action: Oxytetracycline can inhibit the synthesis of the bacterial cell wall, preventing the proliferation of bacteria and stopping the spread of infections (17).

2. Formic Acid:

Description: Formic acid, which is effective against Varroa mites, can be vaporized or dripped into the hive. However, it can cause side effects on bees and requires careful dosing.

Mechanism of Action: Formic acid can affect the respiratory system of varroa mites, causing their death and reducing their population in the bee colony (30).

3. Oxalic Acid:

Description: Oxalic acid, another chemical used in varroa control, can be applied by dripping or evaporation. Correct dosage and timing of application are important.

Mechanism of Action: Oxalic acid can disrupt the acid-base balance of varroa mites' body fluids, causing their death and reducing their population in the bee colony. It is effective in removing mites from adult bees (17).

Natural and chemical substances play an important role in the control of bee diseases. However, their use should be carefully managed and their environmental impact should be considered. Furthermore, beekeepers should adopt integrated bee management techniques to maintain bee health.

In the USA, 11 drugs, each containing 1 of 3 active ingredients (lincomycin hydrochloride, oxytetracycline hydrochloride and tylosin tartrate) have been approved by the Food and Drug Administration (FDA) for use in controlling infections in honey bees. Of these 11 drugs, 10 require a prescription and 1 is subject to the Veterinary Feed Directive (VFD) (17).

In the study investigating the effects of Bayvarol, Fumidil-B and Neo-Ter-mycin drugs used against bee diseases on bee lifespan, 20 colonies were divided into 4 groups and the effects of these drugs were observed. As a result of the study, it was determined that some drugs used against bee diseases and pests had a negative effect on the life span of honey bees (6).

Synthetic acaricides and antibiotics are used to control diseases and mites. The use of pesticides leads to the development of drug-resistant organisms, adversely affecting non-target organisms and causing residue problems in bee products (31,32)

Conclusion

Bees are creatures that have very important roles in the ecosystem. In addition to producing honey, they provide pollination of plants. Bee health is very important for both the natural products they produce and the ecosystem. Today, bee deaths occur due to many different factors. Pesticides, heavy metals,

environmental conditions and biological factors cause bee deaths. Bee diseases are the leading cause of these deaths. For this reason, intensive studies are carried out to prevent bee diseases with natural methods or harmless chemicals. In this section, bee diseases and natural and chemical substances used as preventive against these diseases are mentioned. However, it is seen that more detailed studies are needed to increase the variety of these substances and to determine their possible side effects.

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Chapter

2

USE OF ESSENTIAL OILS OF PLANTS IN CONTROL OF VARROA

Münire TURHAN

Introduction

After the varroa parasite was first seen on honeybees (*Apis mellifera* L.) (1), it quickly spread to all countries of the world and caused great damage to honeybee colonies (2). To date, 20 genotypes of varroa have been identified, and three of them have negative effects on *Apis mellifera*. The parasite, which was identified as *Varroa jacobsoni* by Oudemans for years, was determined to be *Varroa destructor* (2). To date, chemical, biological, genetic and hormonal control methods have been used in various countries against *Varroa destructor*, which survives by sucking the blood of larvae, pupae and adults of honey bees (3,4,5,6,7,8). It is of great importance to use various chemical substances in the fight against *Varroa destructor* and in controlling it (9,10). To date, at least 146 different fumigants, acaricides, insecticides and drugs obtained from plant-derived oils have been tested against this parasite (11).

In recent years, licensed chemical drugs such as Folbex-VA®, Formic Acid plates® (IMP), Perizin®, Antivarroa®, Apitol®, Apistan®, Bayvarol®, Varropol® have

been widely used in varroa control in various countries of the world (12). , 9,13,14 15). However, the varroa parasite's immunity to chemical drugs such as fluvalinate, flumethrin, amitraz and coumaphos, which have been used in recent years, leaving residues in honey and beeswax, and not being effective on varroa in closed nestling eyes, have resulted in some important problems encountered in practice (16,17,18,19,20).). Varroa destructor is still on the agenda as an important problem in the world beekeeping industry today. Many new applicable models and methods have been developed for the control of this pest (21, 22,23, 24, 25).

However, since these methods are quite expensive and require intensive labor, and some of them are not practical and effective, alternative methods, especially the use of various organic acids and aromatic plants, have become widespread. The effect of natural substances and volatile fatty acids extracted from aromatic plants on varroa control has been discussed by researchers (26,27,28), and it is reported that wormwood and cumin plants are used in varroa control (29). Wheat flour (30), menthol (31), sage and thyme oil (32), *Coriandrum sativum* (33), eucalyptus mint and wormwood (34), tymol, menthol, eucalyptol, camphor (35,36) were tested on varroa.

It has been determined that the plants *Allium sativum*, *Nicotina tabacum*, *Juglans regia*, *Lycopersicon esculentum*, *Artemisia absinthium*, *Pinus sylvestris* used in Varroa control are 50-80% effective against this parasite (37). Smoke obtained from a mixture of creosote bush, grapefruit, and cedar leaves has been used to control varroa (38). Varroa control with walnut leaf smoke and pollen traps is effective; It has been reported that varroa control in colonies can be made safer and more effective if this method is applied in combination with physical control methods (39).

The use of oxalic acid and lactic acid against Varroa has been widely used in some countries in recent years because it is effective, simple and low-cost. (40, 41, 42, 14). Unless a method of control that is effective against this parasite and harmless to bee products is found, beekeeping and the plant production sector that requires insect pollination will continue to be under economic threat; Although the extremely negative side effects of chemical substances in terms of bee and human health have been revealed in recent years (17), unless alternative control methods are put forward, the use of chemical substances will continue and their use will not be abandoned for many years and will be discussed (11,43,16, 44).

Integrated control methods used against the Varroa parasite in recent years have reduced the use of synthetic pyrethroids and brought to the agenda the use of various genetic, biological and plant-derived products. Considering that the concept of organic agriculture is becoming more widespread day by day and the difficulties encountered in placing agricultural products in foreign markets, it reveals the importance of the practices in the beekeeping sector. The high level of pesticide residues in honey sold for export creates significant problems in marketing. It is also a fact that this health problem encountered in the export of honey is also valid for the domestic market, but this issue is not emphasized because it is not emphasized much.

In the near future, in parallel with the development level of consumer awareness, such problems are expected to occur in the domestic market. For this reason, beekeeping practices and the use of drugs and chemicals in the fight against varroa, which has an important place in these practices, is an issue that needs to be emphasized. Effective fight against varroa, which is not harmful to bee and human health need to be put into practice, developed and disseminated. In this research, by comparing herbal, biotechnical and chemical applications in Varroa destructor control in honey bee colonies; The feasibility of safe and applicable alternative control methods to reduce intensive chemical use and eliminate the residue problem in bee products has been demonstrated. Herbal, biotechnical and chemical applications in Varroa control; effect levels against varroa, honey bee colonies

Its effects on population development and honey productivity levels of the colonies were determine

1. Varroa Body Structure and Life Length

Adult female varroas are 1.1-1.2 mm long, 1.5-1.7 mm wide, dark red brown in color, male varroas are smaller than females (0.8-1.0 mm), and their body color is pale brown-yellow. Male and female varros are large enough to be seen with the naked eye. The back, covered with a hard chitin layer, has an oval structure. Varroa destructor females are more spherical and larger in shape than Varroa jacobsoni females. The mouth structure of female warros is piercing-sucking. The mouth structure of male varroa has developed to transfer sperm to female varroa. For this reason, male varroas cannot feed and die shortly after mating in their closed brood cells. Varros have four pairs of legs with

six segments. The legs are short and thick, with a number of sensory hairs on them. Female varros can live for 2-3 months in summer and 5-8 months in winter. Reproduction of female varroa begins with incubation activities in the bee colony in spring and continues until autumn. It can survive on adult worker bees without laying eggs during the winter months and the broodless period of the colony (45).

1.1. Development and Proliferation

Varroa is a very dangerous external parasite that feeds by sucking the blood of adult bees and developing larvae and pupae. Their effects are very high during the larva and pupa periods of bees. Adult female varroa larvae enter the eye of the 5-5.5 day old bee larva shortly before the eyes close, feed by sucking the larva's blood before laying eggs, and lay the first egg 60 hours (2-3 days) after the eyes close. Each female varroa can lay between 2-6 eggs every 30 hours, and usually the first egg produces a male and the second egg produces a female. In a worker bee's eye, 3 female varroa can become adults, and in a drone's eye, 5 female varroa can become adults. (45). The eggs laid in the bee's eye go through the 6-legged larva, 8-legged protonymph and deutonymph periods and become adult varroa in 6-8 days.

Females mate in closed eyes and male varros die after mating. Varroas that do not mature in the young cells die inside the honeycomb cell during the emergence of the adult bee. Mature female and newly fertilized female varroa come out of the eye with the adult bee. These female varroa can re-enter the comb cells between 3 and 150 days, depending on the season and the status of offspring in the colony. Females hold on to the head or neck area of the abdominal segments of adult bees, suck the blood of the bee in their area, and share in the food of the bees and larvae. It has the ability to move very quickly within the colony with the help of its legs and body hair. Although they prefer drone eyes over worker bee eyes, they also reproduce in worker bee eyes.

More than one varroa can be seen in one eye in a colony with a lot of contamination (46). To the development and proliferation of Varros; Genetic factors, suitability of colony conditions, amount of brood space, and varroa infestation rate of the colony are the most important factors affecting it. The gender and

race of the larva on which it develops are effective in the reproduction of varroa. The reproduction rate can vary between 1.8-2.9 in worker bee eyes and 2.7-3.7 in drone cells. There is variation in development and reproduction among breeds. The earlier the production of offspring begins in colonies and the later it ends, the faster the reproduction rate and development of varroa species increases.

The presence of a large number of parasite-infected colonies within the flight area of bees, bee confusion, honeycomb reinforcement and the application of ineffective control methods cause other colonies in the apiary to become infected with this parasite (45).

Varroa damage may occur depending on the individual and number of individuals in each larval eye. If the number of varroa in the colony is low, no significant interaction can be seen. The presence of 1-2 varroa in the young eye reduces the life force of the bee that emerges from the eye. The presence of more than two varroa cells in the brood eye has negative physical and physiological effects during the development period of the bees. These effects; shortness of life, loss of wings, shortening of the abdomen, deformations in the wings and forelimbs, pupal deaths, loss of live weight in bees, protein loss in larvae and pupae, decrease in colony development rate and production efficiency, decrease in sperm production in male bees, decrease in flight efficiency of bees, honey and It can be seen as significant decreases in pollen collection efficiency, loss of natural resistance to infections, creating a source of infection and increased winter losses (46).

It is often not easy to determine varroa contamination in colonies. Even if the contamination is intense, negative traces may not be seen until the end of the summer season. Honey yield can be high. On the other hand, upon close examination of the colonies, the number of varroa can be high on adult bees and in male and worker bee cells. In colonies where contamination is intense, a significant decrease in the activity of raising offspring and collecting honey and pollen can be observed (47).

Bee virus diseases are common in colonies with high Varroa infestation and have a negative impact on the colonies. Virus-related infections develop at points where the varroa sucks blood from the bee's body. As a result, Slow Paralysis Virus (SPV), Deformation Wing Virus (DWV) and Acute Bee Paralysis Virus are common in colonies. Although there is no precaution to be taken

against bee viruses, effective control against varroa causes the recovery of virus diseases (47).

Failure to apply any control method against colonies with high Varroa infestation negatively affects the colonies. The level of contamination reduces the regenerative power of adult bees, weakens and disrupts the social order in the colony, and ultimately eliminates that colony. If the number of varroa in the colony is more than 2500, it causes the colony to collapse. Most colonies collapse between August and September, some completely during the winter. Sudden decreases in the adult bee population in the colony, deformations in the wings, legs and abdomen of adult bees, a sudden increase in the number of varroa in the brood cells, irregular and weak brood patterns in the brood, and an increase in the number of dead larvae are the most important symptoms of the collapse of colonies (45).

1.2. Purpose of Varroa Control

It is important to keep the varroa population within the colony at a level where it cannot be harmful. Aiming to eliminate all varroa in the colony with an effective control method should not be the primary goal. The presence of too many varroa in the colony does not affect the faster development of the varroa population (45). It should not be assumed that an effective control method used for Varroa control for one year will be effective and sufficient in other years as well. Continuous monitoring and control of colonies for varroa is important for the future of the colonies. The development of varroa infestation level in the apiary should be monitored regularly in every season and from the information obtained, it should be revealed which method should be used against varroa in which season.

In some years, the level of varroa infestation in bees may develop faster than in normal years, and this infestation development may vary from apiary to apiary (47). This work; It was carried out to find out the effect of varroa on varroa, which has been damaging bee colonies for years, by using various herbal, chemical and biotechnical application methods. This study was conducted to compare and examine the applications in terms of having the least negative impact on bees, not leaving residue in the honey, and being more economical.

2. Use of Essential Oils

Gal et al. (1992), in their study to determine the effectiveness of thyme and mint plants against varroa; Since some of the drugs used leave residues in honey and beeswax, plant-based substances should be applied; They determined that while sticks impregnated with 20-33% origanum oil had an effect of 85-91%, 10% thyme oil caused 54-85% varroa death.

Shaarawi (1995), in his study on the use of some natural substances in varroa-infected colonies; It revealed that there was a 56.82% decrease in the varroa population with eucalyptus leaves.

Calderone and Spivak (1995), in a study they conducted with cineol solution; While they emphasize that cineol has a 96.7% effect on varroa, they report that this value remains at 4.4% in the control group.

Amrine et al. (1996) report that essential oils have a poisoning effect on varroa through direct contact and that feeding bees with syrups containing essential oils weakens the reproductive power of varroa. Application of high doses of essential oil prevents female varroa species from laying eggs on bee larvae, and low-dose essential oil application prevents them from laying eggs; It shows that the varroa species stop their development and the adult bees die inside the honeycomb chamber before they can emerge from the cell and the varroa species become adults.

While the nurse bees feed the larvae, the essential oils in royal jelly are taken in by the larvae. These essential oils, which enter the blood of the larva through food, have negative effects on the reproduction of varroa that feed on the blood of the larva. It is reported that these essential oils used in the fight against Varroa have no negative effects on the reproduction and offspring development of bees.

Barton et al. (1997) states that the formula of cineol is $C_{10}H_{18}O$ and its trade name is Eucalyptol, that cineol and its other compounds do not harm the environment, that it can easily dissolve in soil and is added to various foods, chewing gum and throat lozenges in low doses, that essential oils do not have a toxic effect on humans, and that Thymol is harmful to the eyes. It is reported that wax can absorb essential oils due to its hygroscopic properties, but it dissipates over time. Cineol containing essential oils has antimicrobial properties, eucalyptus oil contains high levels of cineol and They emphasize that it has a pesticide effect.

Kevan et al. (1997), 150 essential oil compounds were used effectively against bee diseases and pests due to their high evaporation properties; They emphasize that these essential oils have a lethal effect of less than 10% on honeybees and are successful in controlling varroa, tracheal mites and foul rot.

Çakmak et al. (2002) stated that walnut leaf smoke application is effective in varroa control; They report that applying this method together with physical control methods is safer and more effective on varroa control in colonies.

Anonymous (2003), essential oils in eucalyptus leaves; rich chemical compounds such as polyphenolic acids, hydrocarbons, ethers, esters and flavonoids states that it exists. Since eucalyptus is an aromatic plant, it has It contains substances that have biological and physiological effects, especially in the leaf part.

It is used for therapeutic purposes in varroa control because it contains large amounts of essential oils. is stated to be used.

Abbadi and Nazer (2003) reported that lavender oil was used to control varroa. In a study using black pepper oil, sage oil, thyme oil and eucalyptus leaves. black pepper oil gave very good results compared to the control group, but between applications They report that there is no significant difference in terms of varroa mortality.

In her study, Çetin (2010) used orange peel, eucalyptus bark and leaves in the form of smoke with a blower in the hive entrance hole in the fight against varroa and found that it was 99% effective in orange peel and 57% effective in eucalyptus bark and leaves.

Turhan (2020) used the leaves, fruit and essential oils of the myrtle plant in the fight against varroa and found very effective results.

Conclusion

Since plants are easy to collect from nature, they will be more effective and economical for beekeepers. It would be beneficial to use these substances because they have positive effects on colony population development and honey yield. It is important for human health that honey and beeswax do not leave any residue. The use of these substances, which are effective enough to compete with perizin, one of the most effective varroa drugs, will also contribute to the availability of quality honey in the market, as it will bring quality in honey.

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Chapter

3

AMERICAN FOULBROOD

Ayşe Ebru BORUM

American Foulbrood Rot (AFB) is one of the most important infections affecting all life stages of honey bee (*Apis mellifera* L.) brood.. It is a infection that can be seen at any time of the year when honey bee brood is present. The causative agent of the disease is *Paenibacillus larvae* (*P.larvae*), a spore-forming. AFB is the most devastating bacterial infection of honey bees (*Apis mellifera* L.) (1). Infection is not only dangerous and lethal for honey bee larvae. Loss of offspring due to increased egg-laying activity in the colony can no longer be replaced by the queen bee. As a result, the population decreases, which can lead to the collapse of the whole colony. The number of adult bees in the colony constantly decreases and the colony strength decreases. (18, 19, 34).

Infected bee colonies can spread the disease between colonies and apiaries. Lack of nectar flow causing predation and improper beekeeping can accelerate the spread. The agent causing the infection is classified as a highly contagious pathogen. Death is inevitable for infected larvae. If the infection is not diagnosed at the early stage and suitable precautions are not taken, it will cause serious losses in infected honey bee colonies (24, 25, 56).

1. Etiology of the Causative Agent of the AFB

The causative agent of AFB is *P. larvae*. It is a Gram-positive, mostly motile, rod-shaped and spore-forming bacterium (Figure 1). In each infected honey bee larva *P.larvae* can produce more than one billion spores. The bacterium is a mostly motile rod-shaping with a straight or sometimes curved, rounded tip. The size of the bacillus varies (0.5–0.8 μm wide and 1.5–6 μm long). It forms filaments and chains.

It was previously named *Bacillus larvae* White (61). *P. larvae* endospores are most commonly found in sticky and dried larvae and old cultures (23, 61, 64).

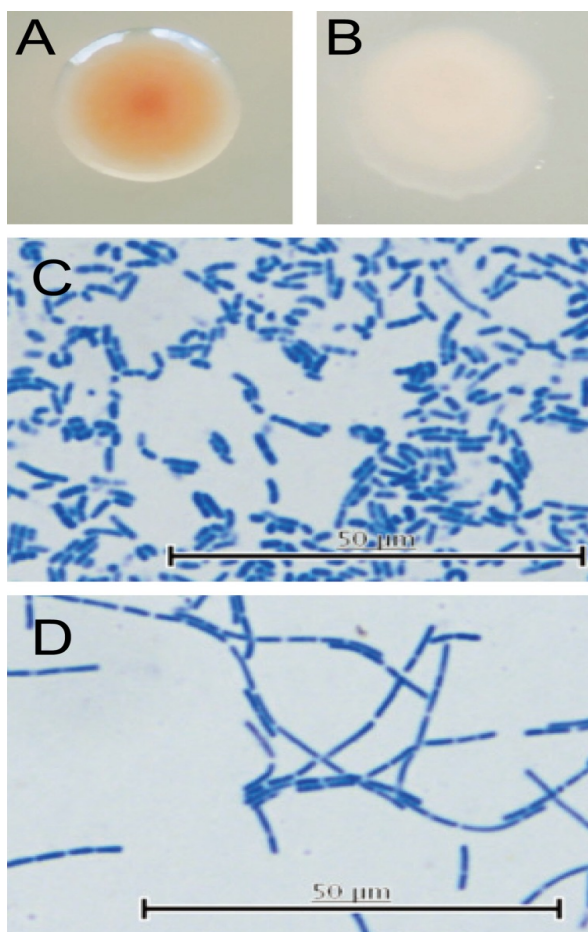


Figure 1: J agar; (A) pigmented and (B) non-pigmented *P. larvae* colonies.

Gram staining method; (C) short rods formed by pigmented ERIC II isolates, (D) long rods and chains formed by non-pigmented ERIC I isolates

(<https://europepmc.org/article/pmc/pmc4976278>).

Over one billion endospores can be found in a single cell of an infected brood comb or a single dead larva (25). Spores are mostly centrally located and elliptical. Spores are highly resistant to high temperatures and many physicochemical agents (13). Dried larvae, old comb remains and endospores remain infectious in the environment and pathological material for about thirty-five to forty years. Spores in, beeswax, propolis, honey and other bee products can remain viable for more than seventy years (29).

Genotyping of *P. larvae* has been performed by molecular methods (23, 41). The most frequently isolated genotypes were reported to be ERIC I and ERIC II. Information on ERIC III and ERIC IV is more limited (25, 30, 41).

In a recent study, a new genotype ERIC V was discovered in isolation from Spanish honey samples. Endospore forms vary in all five genotypes. Spores of ERIC I and II genotypes have a smooth surface, while spores of ERIC III-V genotypes have longitudinal ridges. ERIC I and II correspond to the former subspecies *P. l. larvae*, ERIC III and IV to the subspecies *P. l. pulvifaciens* (3). All of these genotypes are pathogenic for honey bee larvae. The clinical findings they produce are similar. However, colony appearance, pigmentation, haemolysis, carbohydrate fermentation and virulence characteristics of the genotypes are different (3, 13, 23, 25, 30, 31).

Table 1. Characteristics of *P. larvae* genotypes (3, 23, 25, 5, 62).

Characteristic	ERIC I	ERIC II	ERIC III	ERIC IV	ERIC V
Virulence	Death occurs within 12 days	Death occurs within 7 days	Death occurs within 7 days	Death occurs within 7 days	Death occurs within 3 days
Frequency	The most commonly isolated genotype in the world	The most commonly isolated genotype in the world, more common in Europe	Unidentified in recent years	Unidentified in recent years	Isolated and Identified in Spain
Pigmented colonies	-	+	+	Variable	-
Spore surface	Smooth	Curly	With ridges	With ridges	With ridges

Growth in nutrient broth	-	+	+	+	Not Tested
Mannitol Fermentation	-	+	+	+	Not Tested
Salicin Fermentation	+	-	-	-	Not Tested
Alkaline phosphatase	+	-	+	+	Not Tested
Acid phosphatase	+	-	+	+	Not Tested
Catalase	-	-	Weak, delayed +	Weak, delayed +	Not Tested

Virulence properties of different genotypes of the pathogen are different. Generally, larvae infected with ERIC I genotype die within 10-12 days, larvae infected with ERIC II-IV genotypes die within 6-7 days and larvae infected with ERIC V genotype die within 3 days. (3, 25, 50, 57).

In colonies infected with ERIC II genotype, characteristic clinical findings are observed more rarely. Therefore, clinical examination may give false negativity (31, 51).

Epidemiological studies have shown that infected colonies most commonly contain the ERIC I and ERIC II genotypes. Among disease cases, ERIC I is the genotype that is most frequently isolated, while the occurrence of ERIC II is comparatively restricted (10, 22).

Spores of the agent are highly resistant to high temperature and different chemical substances. AFB infection is primarily caused by the spore form of the bacterium. The most effective means of transmitting the infection to larvae is through nurse bees or spores present at the base of the brood cell. All larvae, including those of worker bees, drones, and queen bees, are susceptible to AFB infection. However, it is rare for queen bee and drone larvae to get infected in natural conditions. The susceptibility of larvae to AFB disease decreases as they grow older (10, 40). Larvae that are more than 53 hours old cannot be infected. The average dose required to initiate infection, known as LD50 (the spore dose at which 50% of the larvae are killed), typically ranges from 8.49 ± 1.49 spores for honey bee larvae aged 24-28 hours, although this can vary (27)..

2. Transmission and Spread

Dried larvae containing large numbers of spores, diseased and dead larvae are the primary sources of infection. Young worker honey bees trying to remove larvae, honey, royal jelly and pollen from the hive, comb, frame and inner walls of the hive contaminate the colony with the organism. A contaminated comb cell can contain more than one billion endospores. Every comb frame with diseased and dying larvae is a major cause of the spread of infection (24, 34).

The LD50 required to initiate infection is generally 10 or fewer endospores for a 12-36 hour old honey bee larvae. However, it may vary (10, 25).

Infection develops when larvae are fed food contaminated with *P. larvae* endospores by nurse bees. While cleaning the dead larvae, the keeper bees spread the spores of the agent throughout the hive. Hygienic behaviour of bees can also sometimes cause the spread of the disease (13, 23).

P. larvae spores have the potential to maintain their viability for a period of 3-10 years in bee products and the surrounding environment. In the case of dried larval remains, the spores can remain viable for up to 35 years. However, purified spores exhibit remarkable longevity, with the ability to remain viable for more than 70 years. The clinical symptoms of the disease may vary depending on factors such as the specific genotype involved, the strength of the honey bee colony, the stage of the disease, and the colony's resistance to AFB. Therefore, different clinical findings may be seen in AFB infection (22). The larvae are found curled up at the base of the uncapped brood cells, where they rapidly perish during the early stages. The adult worker bees take care of removing the deceased larvae by emptying the cells. Once an AFB-infected larva dies and settles at the bottom of the brood cell, it undergoes drying and hardening. The presence of a black, flat, charcoal-like substance adhered to the bottom of the brood cell is a distinct symptom of AFB. If the larvae are infected with the ERIC I genotype, they typically succumb to the infection after the incubation cell is sealed. On the other hand, larvae infected with other genotypes die prior to cell capping (5).

As the age of the larvae increases, their resistance against infection increases. Endospores develop in the intestines of the larvae in about 12 hours. The bacterium colonises the middle intestine (64). *P. larvae* produce numerous virulence factors such as SplA, C3larvin, PlCBP49, Plx1, Plx2, which cause damage to the intestinal epithelial barrier (14, 21, 49). These virulence factors enable *P. larvae* to adhere to the cells of the midgut. They also lead to ADP-

ribosylation of DNA, chitin degradation, inflammation and cell death without apoptosis (14, 21, 49). It is difficult for worker honey bees to clean infected larvae. Infected larvae are a constant source of infection for other new brood in the colony. The agent can spread between colonies through robbing, drifting, transfer of brood frames and honey frames, tools, equipment, swarming, feeding contaminated honey and pollen and beekeeping activities (18, 19, 58). Honey bees that ingest *P. larvae* endospores in bee faeces and honey can transmit the pathogens (44, 58). The disease can be transmitted by foraging activities of bees, use of unsterilised beekeeping tools, bee excrement, buying and selling uncontrolled bee colonies. If the bees are not provided with the clean and high quality water they need, it is also important in the emergence of the infection that the bees meet their water needs from puddles and pools in the environment that are at risk of contamination with *P. larvae* spores (25). Insects such as ants, flies and wax moths are also effective in spreading the disease (42).

Although adult bees are not infected with the disease, colony losses occur because bee colonies cannot rear offspring (24). As a result of the progression of the disease, old bees in the colony die and new and young bees are not formed in the population. As a result, the honeybee colony continuously weakens and dies. In these colonies robbing and wax moth infestations are common. This often causes the infection to spread further (13, 24, 36, 53). **3. Clinical Findings**

Clinical AFB symptoms are variable. The genotype infecting the colony, the strength of the colony and its resistance to AFB are effective in the occurrence of different clinical symptoms (13, 50, 59). Clinical symptoms may not be observed most of the time. Hygienic nurse honey bees remove dead and diseased larvae while cleaning the brood cells. This behaviour is often seen in colonies infected with endospores of ERIC II-V (23, 64).

Although AFB affects only the larvae, it also weakens the colony. It can lead to rapid death of the colony within three weeks (38).

The color, consistency, and structure of the larvae change soon after the brood cells are sealed. Among healthy brood cells, dispersed brood cells are frequently seen. Whole brood cells, empty cells, uncapped cells, and cells with the remains of diseased larvae are all found together in severely affected colonies. Such combs have an irregular and patchy brood pattern. After the changes in the larvae, the first clinical signs of the disease appear in the colony. However, before clinical symptoms are observed, differences in brood cell caps are remarkable. Usually about twenty

days after infection by larvae, changes in the colour, arrangement and integrity of the brood cells are observed. After changes occur in the cell caps, the typical hygienic behaviour of bees appears. Infected and dead larvae are impacted by the microclimate within the hive. The cell tops turn dark brown after first turning lemon-colored. Cell caps are moist, squishy, and somewhat concave (23, 37, 51).

In autumn, when the queen honey bee stops laying eggs, only infected young remain in the comb, so that the changes in the cell caps are more pronounced. When the brood cells are uncapped, the larvae change colour from pearl white to a distinct bright colour. The colour changes first to sand colour, then to light brown and finally to dark brown (chocolate colour). Infected larvae lose their distinctive form. Their bodies take on a semi-liquid, sticky, thread-like appearance (13). At the onset of the disease, the larvae grow 5-10 cm like a thread when touched with a toothpick or match (Ropiness test). The Ropiness test is the most well-known approach for diagnosing infection in the field, and it may be used under field conditions.

However, sometimes larval remains are quite watery and cause the matchstick test to be negative. Approximately 1 month after the larva has elongated like a rope, the diseased larval remains dry out. These remains adhere strongly to the underside of the cell, forming the brittle and typically hard, dark scales. The decomposition process of the larva takes 5-8 weeks. In the last stage of the disease, the larvae dry out and the larval remains adhere tightly to the cell wall. It has the appearance of a black and brown scale. Rarely, dead larval remains may be watery and have non-specific content (9) (Figure 2).

The mortality rate of larvae infected with *P. larvae* is dependent upon the genotype of the larvae and occurs within 3-12 days (3, 22-25). The ERIC I genotype exhibits lower lethality than other genotypes, resulting in later death of larvae infected with the ERIC I genotype. Larval mortality occurs some time after the brood cells are capped and metamorphosis begins (25, 30, 43).

New spores are formed after larval death. Larval death usually occurs after the cell is capped, either in the last two days of the larval stage or the first two days of the pupal stage. If nurse honey bees cannot remove the larvae from the cell, or if the bacterial spores dry out, they form a scale adhering to the cell. This appearance is typical for the disease (Figure 3) In the advanced stage of infection, a bad odour is felt when the hive is opened. Infected hives may exhibit a sulphurous odour, which may be indicative of the presence of rotting brood (38).

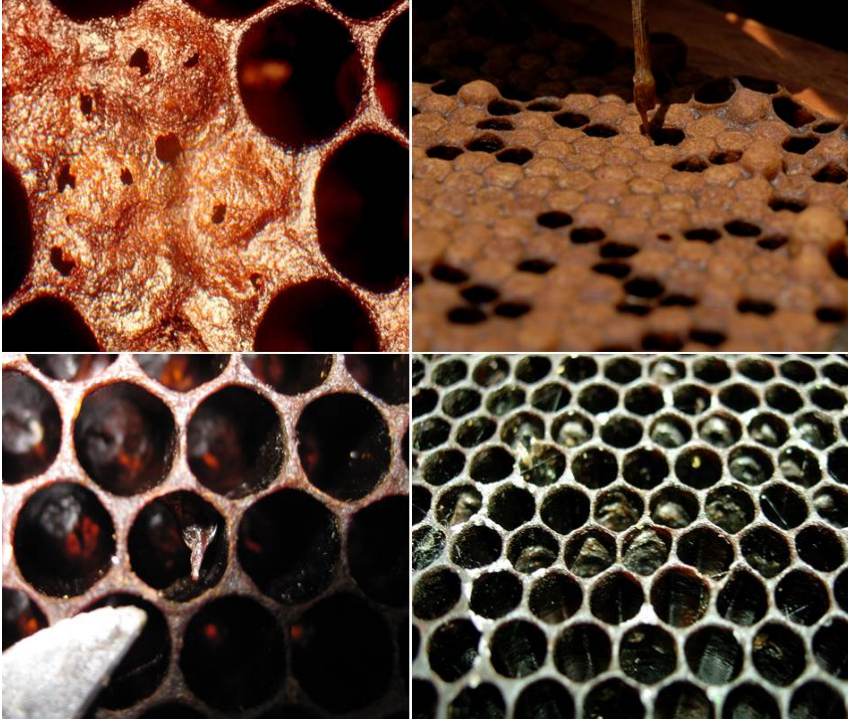


Figure 2: The following images illustrate the symptoms of American foulbrood in a honeybee colony. From top to left, going clockwise: punctured brood cappings, ropy pupae, scales in cells, pupal tongue.

<https://blogs.ifas.ufl.edu/entnemdept/2016/12/23/new-year-new-rule-look-fda-ruling-prescriptions-honey-bees/>



Figure 3: Perforated cappings, and scales

<https://beeinformed.org/2013/10/21/american-foulbrood-afb/>

From infected brood, the characteristic odour of volatile compounds is emitted, which increases the honey bee cleaning instinct. Detection of infected brood is mainly done by olfaction. The presence of valeric acid, propionic acid, isovaleric, butyric, 2-methyl-butanoic, hexanoic and 2-onanone has been reported in infected larvae by Gas Chromatography-Mass Spectrometry (GC-MS). These volatile compounds were not reported in healthy larvae (33, 26).

Nurse honey bees open the cell caps and remove dead and infected larvae. ERIC II-V genotypes are more virulent. At the onset of metamorphosis, infected larvae are removed and killed before the cells capping. Infected larvae are detected by the nurse bees at an earlier stage and removed from the healthy brood. The hygienic nurse bees of the hive are able to remove these larvae in a relatively short period of time. Thus, the vegetative forms of bacteria in the colony are sporulated less. Thus, the agent spreads more slowly within the colony. The virulence of ERIC I is lower than the virulence of ERIC II (35, 43).

In colonies infected with the ERIC I genotype, larvae and pupae in the capped brood cell die (30). In the event of infection of older larvae, the 'tongue' of the pupae may become adhered to the upper surface of the cell wall. Although this clinical finding is rare, it is quite characteristic for AFB (Figure 4). European foulbrood should be considered as a differential diagnosis. In colonies infected with ERIC I genotype, the rate of elongation of larvae into threads is higher. In colonies infected with the ERIC II genotype, only a few cells may be infected, and extension may not be apparent. Consequently, a false negative result may be obtained (9).



Figure 4: A honey bee pupa with American foulbrood disease
<https://afb.org.nz/pupal-tongue/>

4. Diagnosis

The clinical symptoms of AFB can be confused with European foulbrood, sacbrood and varroosis, which are mostly seen in early spring (9, 45, 46). Sacbrood infection is a viral disease of fry. Unlike AFB, the dead grey-brown larvae take the shape of a bag. In early spring, together with varroosis, the death of honey bee brood in capped cells and the developmental forms of pupae, young honey bees deformed, can be seen. European foulbrood affects both capping and uncapping brood cells. Therefore, in EFB, larvae die in both open and closed brood cells (39, 45).

In the case of Varroosis, a pupa or young honey bee is usually found in the cell with one or more *Varoa* mites after the caps are opened. In AFB, however, there is no stage at which bee development is complete. The larvae has dried up and the brood cell appears empty. However, on closer examination, black and brown scales can be found underneath the cells. This finding is common in AFB cases older than two months (39).

In a bee colony with suspected AFB, it is sufficient to send a honeycomb sample containing brood cells suspicious for infection to the laboratory. The whole suspected comb should be sent for examination. If it is not possible to send the whole comb, a part of at least 20 cm² containing diseased and dead larvae with clinical signs may be sent. When laboratory diagnosis of AFB is confirmed in suspected colonies, it should be considered that other hives in the vicinity of the infected hive may also be infected. For definitive diagnosis, samples of honey, beeswax, royal jelly, pollen, bee food, adult bees and combs with young should be taken from these colonies and sent to the laboratory (9, 32, 45, 52, 60).

Brood comb honey is important in early diagnosis. *P. larvae* spores in brood comb honey contaminate the mouthparts of nurse honey bees. Even if first-instar larvae feed only on glandular secretions during this period, infection occurs during feeding. It has been demonstrated that nurse bees that encounter the first foulbrood scale and attempt to remove it become contaminated with *P. larvae* spores and subsequently carry them to the honey of the brood comb. It is therefore evident that brood comb honey is a highly sensitive method for the early detection of AFB. It is stated that brood comb honey is suitable for the early detection of colonies that are infected but do not yet show clinical signs

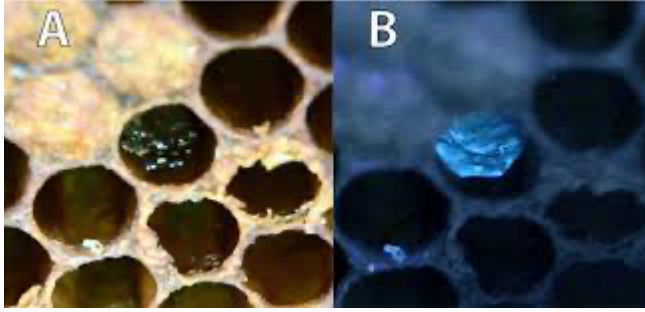


Figure 5: Hard scale using normal light (A) and a black light (B).
Images by Stephen J Repasky

(15). Adult bees in infected colonies can carry many spores. The use of adult bees as a matrix for the detection of *P. larvae* in infected honey bee colonies has been demonstrated to be a highly effective method. (35).

In the event that *P. larvae* are isolated in samples sent from colonies suspected of exhibiting symptoms of AFB, the colony is considered to be infected. Nevertheless, the detection of *P. larvae* spores is not a definitive indicator of the occurrence of an AFB outbreak in the colony (20).

It is not possible to visually examine colonies in cold weather conditions. It has been suggested that winter hive debris may be useful for the detection of AFB-infected colonies, particularly in the spring. Hive debris can be collected from the bottom board. The most significant advantage of utilising hive debris for sampling is that the sample can be obtained without the necessity of opening the hive and disturbing the colony. One disadvantage of using hive debris for sampling is that it may contain a number of environmental, saprophytic bacterial and fungal species that may inhibit the detection of *P. larvae* (6, 15).

AFB-affected scale glows when u.v. black light is shone. The light is shone on the frame. Scale glows with a greenish-bluish colour (38). (Figure 5).

4.1. Field Diagnosis Tests

4.1.1. Ropiness test

Brood cells containing clinically suspicious larvae or pupae can be diagnosed for AFB using the ropiness test. To perform the test, a small board is brought into contact with the suspected larva or pupa and stirred. If AFB is present in

the suspected larva/pupa, the residue is pulled outwards in a fine thread 10-30 mm long. The thread is of uniform colour and consistency. It will 'retract' when the thread breaks. However, the ropiness test is not always a definitive diagnosis for AFB. Larvae, pupae and drying larvae in the first stage of infection may not show prolongation. In addition, completely dried diseased larvae or pupae will not be able to come out because there will be no liquid left in the cell (45, 56).

4.1.2. Holst Milk Test

The test is based on the high level of protease produced by sporulating *P. larvae*. A solution is prepared by mixing 1 teaspoon of milk powder with 100 ml. water. Brown pupa samples are collected. It is kept for 1 hour in a warm environment. At the end of the time, light brown appearance in the tube containing the mixture is considered positive (Figure 6) (38).

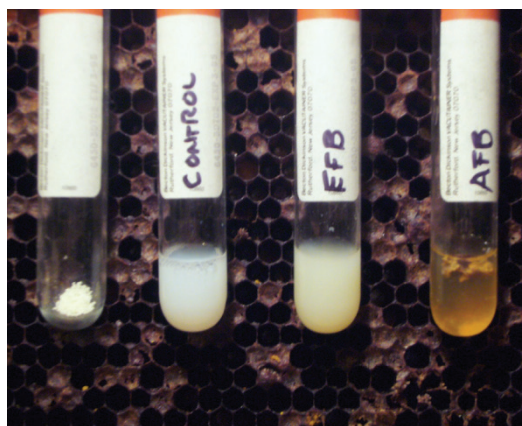


Figure 6: Holst Milk Test

<https://scientificbeekeeping.com/sick-bees-part-18a-colony-collapse-revisited/>

4.1.3. AFB Ready Diagnostic Test Kit

There are commercially available test kits. Brown pupae samples are collected and placed in the extraction bottle provided with the kit. The bottle is closed and shaken for 20 seconds. Put the mixture into the specified section of the test kit. A line under the letter 'C' and 'T' indicates a positive result. Only if a line appears under the letter "C", this indicates a negative result (Figure 7) (38, 45, 56).

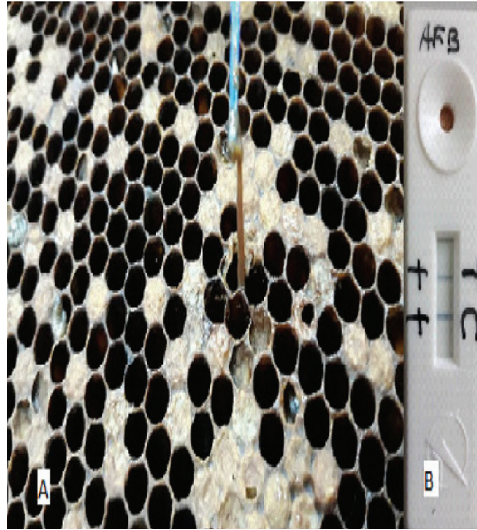


Figure 7: Holst Milk Test

<https://scientificbeekeeping.com/sick-bees-part-18a-colony-collapse-revisited/>

4.2. Laboratory Diagnosis

4.2.1. Culture

Classical microbiological identification techniques include microscopic observation of the spores of *P. larvae*, along with microbiological cultivation and culture. Biochemical tests are the most commonly used methods for the identification of suspected isolates (9, 45). A variety of enriched selective media are available for the culture of *P. larvae*. The following media are recommended for the culture of *P. larvae*: MYPGP, BHIT, J-agar and Columbia agar with sheep blood agar (CSA) (41).

A suspension containing spores of *P. larvae* should be prepared for isolation. To eliminate vegetative forms of other microorganisms, the suspension is subjected to a heat treatment at 80°C for a duration of 10 minutes or 95°C to 96°C for a period of three to five minutes. It is possible to coat larval remains directly on agar without heat treatment (9, 10, 45).

A sterile swab is used to collect larvae/pupae remains from combs with brood. The swab should be suspended in a suitable medium, typically 2-10 ml of sterile water or physiological solution, such as 0.01% PBS or 0.9% NaCl. Add 2-3 drops of sterile water on the larval or pupal remain samples sent by glass

slide. Emulsify with a sterile swab. A suitable medium should be prepared and inoculated with an extract that has been emulsified with the requisite material. The cultures are incubated at 5-7% CO₂ for 7 days and controlled every day (9, 45).

Colonies can be seen from the 2nd day.

It is recommended that the spore suspension be cultured in triplicate.

- a) No heat treatment of the suspension
- b) Apply heat treatment at 80°C for 10 minutes,
- c) Apply heat treatment at 95°C for 3 minutes,
- b) and c) are applied to kill vegetative forms of other microorganisms.

Bacillus, *Paenibacillus* and *Brevibacillus* spp. can grow on semi-selective media with added nalidixic acid and pipemidic acid. In non-heat-treated procedures, amphotericin B at 16.8 µg/ml can be used in addition to both antibiotics to prevent fungal contamination. The plates are incubated at 37±1°C for 2-4 days (9, 10, 45).

To prepare the smears from the dead larvae, add 1-2 drops of sterile water and mix on the slide. A new smear is prepared for microscopic examination of the spores and Gram staining.

ELISA, fluorescent antibody technique (FAT) and immunodiffusion test are antibody-based techniques used for diagnosis (9, 45).

Honey can also be sent as a sample. Honey samples are heated to 45-50°C and the spores are dispersed by shaking. Each honey sample is diluted 1:1 with 0.01 M PBS or 0.9% NaCl (pH 7.2). The resulting solution is then centrifuged at 6000 g for 40 minutes. The supernatant is removed, and about 3 mL is added to each tube. The pellet is resuspended by vortexing for 1 min. The samples are vortexed again at high speed for 2 minutes and 100-200 µl precipitate-liquid mixture is inoculated into the appropriate culture medium to which antibiotic is added and incubated at 37°C for 7-8 days (10).

One gram of pollen was suspended in sterile distilled water or 0.01 M sodium phosphate buffered saline (pH 7.2; final concentration, 10 ml). Then the filtrate is prepared by filtering through Whatman No. 1 paper (10).

Adult bee samples are homogenised by adding sterile water or physiological solution. The homogenate is then filtered through Whatman No. 1 paper.

Subsequently, the pellet is subjected to centrifugation and resuspension in phosphate-buffered saline (PBS). The samples are then heated. The samples are cultured under appropriate conditions (10, 45).

4.2.1.1. Culture Mediums

MYPGP agar

The concentration of MYPGP agar in a litre of solution is as follows: The composition of the medium is as follows: Mueller-Hinton broth (10 g), potassium dihydrogen phosphate (K_2HPO_4) (3 g), sodium pyruvate (1 g), glucose (2 g), yeast extract (15 g), and agar (20 g). Additionally, 9 µg/ml nalidixic acid and 18 µg/ml pipemidic acid are added to the medium.

J-agar

The cost per litre is as follows: The composition of the medium is as follows: 15 g yeast extract, 3 g K_2HPO_4 , 5 g tryptone, 2 g glucose, 20 g agar. Additionally, 9 µg/ml nalidixic acid and 18 µg/ml pipemidic acid are incorporated into the solution.

CSA

The Columbia blood agar base is prepared with 5% sterile defibrinated blood. Additionally, 9 µg/ml nalidixic acid and 18 µg/ml pipemidic acid are incorporated into the medium.

BHIT agar

Brain-heart infusion agar (47 g per litre) is prepared by adding a sterile solution of thiamine hydrochloride (final concentration of 1 mg per litre) to the medium.

MYPGP agar is a standard medium for the diagnosis of AFB. It is the optimal medium for the cultivation of *P. larvae*. J-agar, BHI and CSA are less effective than MYPGP agar. The medium is incubated for a period of 2-4 days at a temperature of $37 \pm 1^\circ\text{C}$ in an atmosphere containing 5-10% CO_2 , under aerobic conditions.

Honey samples should be incubated for 6-15 days and controlled for suspicious colonies on days 3 and 6 (9, 10, 45).

4.3. Colony Morphology

In J-agar and MYPGP, the colonies are small, smooth, mainly rough, raised or regular and white to beige in colour (Figure 8).

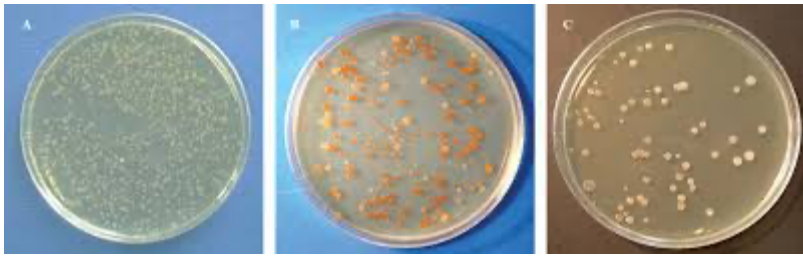


Figure 8: Colonies of *P. larvae*

<https://www.tandfonline.com/doi/pdf/10.1080/01652176.2014.993095>

The colonies that grow on the CSA medium are small, regular, rough and greyish in colour, with a slight transparency and a slight shiny. *P. larvae* form colonies with colours ranging from orange to red. On MYPGP, J-agar and CSA, ERIC II and III genotypes form orange or reddish colonies. The ERIC II and III genotypes are responsible for the formation of pigmented colonies. In contrast, the ERIC IV genotype is responsible for both pigmented and non-pigmented colonies. The morphological characteristics of a bacterial colony are not sufficient for a definitive diagnosis. Nevertheless, it permits the selection of bacterial colonies for subsequent identification (9, 10, 45) (Table 2).

Table 2. The phenotypic characteristics exhibited by the ERIC genotypes of <i>Paenibacillus larvae</i> .. (data based on 3, 22).					
Characteristic	ERIC I	ERIC II	ERIC III	ERIC IV	ERIC V
Pigmentli koloni	–	+	+	Variable	–
Spore surface	Smooth	Convolutad	With ridges	With ridges	With ridges
Growth in nutrient brot	–	+	+	+	+
Fermentation of mannitol	–	+	+	+	Not Tested
Fementation of salicin	+	–	–	–	Not Tested
Alkaline phosphatase	+	–	+	+	Not Tested
Acid phosphatase	+	–	+	+	Not Tested
Catalase	–	–	Weak, delayed +	Weak, delayed +	Not Tested

4.4. Microscopic examination

4.4.1. Gram Staining

Gram staining is usually done to examine isolated colonies of bacteria. *P. larvae* are Gram-positive rods exhibiting slightly rounded tips. Gram staining is performed on larval smears prepared from clinically suspected colonies. *P. larvae* appear as pink coloured ellipsoidal spores (approximately $0.6 \times 1.3 \mu\text{m}$) (9, 10, 45).

4.4.2. Schaeffer and Fulton method for staining endospores

In this method, the spores of the agent appear green, while the vegetative cells appear red (9, 10, 45).

4.4.3. Nigrosin negative staining

The bacterial suspension is mixed with nigrosin solution (5%). In this method, the background is black, while the spores of *P. larvae* appear bright purplish-red and the vegetative forms appear greyish (9, 10, 45).

4.5. Mass spectrometry

In this method, specific mass spectra of bacterial ribosomal peptides are determined. The bacterial species is identified by comparing the detected spectra with a known database. The method is based on a physicochemical technique, namely matrix-assisted laser desorption ionisation time-of-flight mass spectrometry (MALDI-TOF MS) (55).

4.6. Biochemical tests

Catalase test

Different genotypes of *P. larvae* are catalase negative or weakly delayed positive (9, 10, 45).

Carbohydrate tests

The bacterium *P. larvae* is capable of producing acid through aerobic respiration, utilising both glucose and trehalose as substrates for fermentation. It does not produce acid from arabinose and xylose. Mannitol and salicin tests give variable results (9, 10, 45).

Casein hydrolysis

Twenty grams of agar and ten grams of yeast extract were prepared, and thirty millilitres of ultra-high temperature (UHT) skim milk and one-point-five millilitres of filtered 0.1% thiamine solution were added to seventy millilitres of sterilised, cooled medium. *P. larvae* is capable of hydrolysing casein (9, 10, 45).

4.7. Molecular Techniques

Molecular techniques are used for the identification of AFB agent. Pulsed-field gel electrophoresis (PFGE), Polymerase chain reaction (PCR) and real-time polymerase chain reaction (real-time PCR), are the most commonly used molecular techniques. Multilocus sequence typing (MLST), high throughput sequencing (HTS) and multiple-locus variable-number tandem repeat analysis (MLVA) methods are also applied (4, 12, 13, 25, 30, 32, 44).

PCR method is applied for rapid and definitive diagnosis of AFB. In order to prepare samples for PCR, larval remains with clinical findings are emulsified in 1 ml of sterile distilled water and thoroughly mixed. A 100 µl suspension of this mixture is then diluted with 900 µl of distilled water and vortexed. Finally, 100 µl of the mixture is used for DNA extraction by heating and centrifugation. DNA extraction can also be performed using commercial kits (9, 10, 45, 46).

There are various PCR protocols to identify *P. larvae*. Nevertheless, procedures based on the 16S rRNA gene have been demonstrated to be both accurate and reliable (9, 10, 45, 46) (Table 3).

Table 3. <i>P.larvae</i> primers			
Primers	Sequence (5' → 3')	PCR-product size	
AFB-F AFB-R	CTT-GTG-TTT-CTT-TCG-GGA-GAC-GCC-A TCT-TAG-AGT-GCC-CAC-CTC-TGC-G	1096 bp	(Dobbelaere et al., 2001)
PL1 PL 2	AAG-TCG-AGC-GGA-CCT-TGT-GTT-TC TCT-ATC-TCA-AAA-CCG-GTC-AGA-GG	973 bp	(Govan et al., 1999)
ERIC1R ERIC2	ATG-TAA-GCT-CCT-GGG-GAT-TCA-C AAG-TAA-GTG-ACT-GGG-GTG-AGC-G	Several amplicons	(Versalovic et al., 1994)
BOXA1R	CTA-CGG-CAA-GGC-GAC-GCT-GAC-G	Several amplicons	(Versalovic et al., 1994)
MBO-REP1	CCG-CCG-TTG-CCG-CCG-TTG-CCG-CCG	Several amplicons	(Versalovic et al., 1994)

The recent development of real-time PCR methods has enabled the rapid and reliable detection of *P. larvae* from a variety of honey and hive samples. Additionally, larvae and adult honey bee samples can be detected by real-time. Compared to conventional PCR, this method is reliable and sensitive. Unlike other PCR protocols, it can be performed in a shorter time and the risk of contamination is low (46).

Microscopic examination to identify *P. larvae* spores is less sensitive than microbiological isolation. PCR techniques are more sensitive than microscopic examination and microbiological isolation (3, 8, 17). The existence of the endolysin cell-binding domain (CBD), which binds *P. larvae* (54) may be effective in the development of new methods in AFB diagnosis. The effectiveness of applying hyperspectral technology for rapid detection of AFB disease is being investigated (63).

5. Treatment

Early detection of AFB is the first and most important step in controlling and preventing the disease. For early detection of AFB, the colony should be carefully and thoroughly checked in spring. When signs of disease appear, all beekeeping tools and equipment used during the inspection of each colony should be disinfected. Disposable gloves should be used for inspection of sick or suspected colonies and then disposed of. Colony checks should be carried out more frequently and carefully in new cases of disease. If detected in one colony, all colonies in the apiary and colonies within a radius of 3 km near the apiary should be examined. In case of infection, official institutions should be informed (36).

Treatment of AFB is not possible. In accordance with legal procedures, radical measures must be taken in case of disease. All infected hives must be destroyed. After all honey bees enter the hive, the hive entrance is closed and infected colony bees are killed with insecticide. Honey, honey bees as well as combs belonging to the infected colony should be burnt (Figure 9). All destroyed materials and dead bees should be buried. A thorough disinfection is also applied to the treated area (16, 45).



Figure 9: Burning an infected hive and frames

<https://teara.govt.nz/en/photograph/15774/burning-an-infected-hive>

<https://beeaware.org.au/code-of-practice/managing-pests-and-diseases>

After a thorough mechanical cleaning, the hives are burned with a heat source. They are then disinfected with a 2-6% NaOH solution (caustic soda), Na_2CO_3 , alkaline hypochlorite solution or KOH (5% NaClO in the above solutions). Heating to 80°C can be applied. In water containing these disinfectants, the hives are immersed for 5-15 minutes, then washed with warm water and dried. After mechanical cleaning, disinfection of tools and equipment with a solution of 1part Na_2CO_3 and 5 parts water or 0.5% hypochlorite solution (NaClO) for twenty minutes is effective. All tools and equipment should be scraped to remove beeswax and propolis before disinfecting, then scrubbed with hot soapy water using a stiff brush. The wash water should be discarded, and the residues should be burned out of the bees' reach. It is also recommended to disinfect the apiary with a 10–20% CaOH_2 solution (16, 53).

Apiaries and colonies with AFB should be kept under constant control for two months, and attention should be paid to the disinfection of tools and equipment. If no clinically suspicious new AFB colonies are formed within this period and no disease is diagnosed in suspected colonies after laboratory examination, it is accepted that there is no infection in the apiary (7).

In cases where the infection is not at a very high level or at the beginning, shook swarm method is applied in some countries. It is applied by taking into account that AFB is a disease of young honey bees, not adult honeybees. It is a beekeeping procedure in which all honey bees in an infected colony are shaken from their frames and transferred to a clean hive with new foundation comb. All combs with brood, wax, honey, pollen, propolis in the colony are burnt. Adult bees are transferred to a clean hive. The hive can be supplied with combs with young from healthy and strong colonies. Honey bees are fed with fresh sugar syrup (36).

Since the spores are heavily concentrated in brood frames and hive material, the adult bees and queen bee in the colony are moved to new, clean hives and given new foundation combs. Thus, it is suggested that the spore load at the colony level can mostly be reduced (11).

This method can be applied in two different ways. In the ‘partial shook swarm’ technique, all combs with brood are replaced with new foundation combs, while combs with honey and pollen are placed in the hive. This technique allows the colonies to regain strength more quickly than the ‘full shook swarm’ technique, where all combs are replaced. Contaminated combs are destroyed by burning (48).

Firstly, AFB is suspected by looking at the clinical symptoms of the disease in the colony. However, the definitive diagnosis of AFB is made by laboratory diagnosis of the causative agent (25). Colonies and apiaries where AFB has been diagnosed by laboratory testing should be subject to clinical investigation and control measures should be in place (13, 36, 53).

Beekeepers should understand that the use of antibiotics cannot substitute good beekeeping practices (GBPs). GBPs are integrated activities to achieve best health for people, honey bees and the environment. Antibiotics cause residues in hives and bee products. They are also ineffective because the causative agent is spore-borne. Although it temporarily improves clinical symptoms, infection is seen again because it is not effective on spores (53).

If colonies do not show clinical AFB symptoms, an official AFB outbreak cannot be declared in most countries. In this case, methods such as ‘shook swarm’ are recommended to treat colonies that are not yet diseased (28).

6. Prevention and Control

It is important to carry out detailed and comprehensive controls of bees and bee colonies in terms of adult and brood diseases. In addition, appropriate

samples should be taken from the colonies at least twice a year, especially in autumn (before wintering) and spring (before bees start foraging) periods, and laboratory diagnosis should be made (36, 44).

Good beekeeping practices (GBPS), good hygienic practice (GHP) and biosecurity practices are especially important in protection and control. These practices are general and protective procedures to prevent the spread of infection to healthy colonies and apiaries. It is an important and legal obligation to take the necessary precautions in colonies where AFB infection is detected or suspected (39).

Practices such as hive hygiene, regular colony checks, comb replacement, disinfection, controlled queen and colony purchases, sterilised base comb produced where possible using HACCP (Hazard Analysis and Critical Control Points), controlled use of supplementary feed and attention against swarms of unknown origin should be implemented (16, 35, 36, 39, 53).

The health status of hives or colonies of honey bees should be controlled before starting work in the apiary. Every difference in the brood, combs or apiary should be taken seriously by the beekeeper. Only laboratory analysis can provide a definitive diagnosis of suspected diseases in bees and brood, particularly AFB. Laboratory diagnosis of material from any suspected AFB hive is essential (2, 9, 56, 63).

Attention should be paid to issues such as preventing robbing, honeycomb and brood comb transfer between colonies, replacing old honeycombs every 2-3 years at the latest, buying tools and equipment from reliable places, swarms of unknown origin, providing queen bees and colony bees from reliable suppliers.

Good beekeeping practices should be generalised.

- Beekeeping in adequate areas and suitable lands
- If possible, the apiaries should be disinfected before the bees are placed.
- Using sterilised basic frame
- Laboratory analysis to determine the presence of *P. larvae* spores in foundation combs
- Keeping beekeeping records
- Use strong colonies, provide colonies with high quality food
- Ensuring good and sufficient air circulation in the hives
- Use of young, fertile and healthy queens

- Providing clean and quality water to bees
- Not combining weak and strong colonies (sick and healthy)
- Regular inspection of bee colonies
- Provision of quality and reliable replacement honeycombs
- Control of bee colonies when bees are foraging
- Application of appropriate disinfection to tools and equipment in apiary controls
- Taking measures for early detection of the disease
- Controlling stress factors in bees
- Laboratory diagnostics
- AFB-infected honey bee colonies were destroyed through burying and fire.
- Implementation of hygiene and sanitation in the apiary in case of the occurrence of diseases

AFB is a very important and rapidly spreading infection. Early detection of infection and taking necessary precautions are very effective and important in the prevention, control and spread of the disease.

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LABORATORY DIAGNOSIS OF VIRAL HONEY BEE DISEASES

Metin GÜRÇAY

Introduction

Beekeeping is widely practiced around the world and has always had an important place in agricultural activities. The suitable climate and rich vegetation are very suitable for beekeeping. Humanity attaches special importance to beekeeping and honey. The fact that beekeeping activities started in ancient times with humanity proves this. Türkiye is one of the important countries in terms of the number of honey bee colonies in the world. Honey yield per colony is approximately 15.9 kg. However, the honey yield per colony in countries with modern beekeeping activities is 74 kg in Australia, 57 kg in Canada, and 33 kg in America. This situation shows that beekeepers, veterinarians, agricultural engineers and biologists need to work much more on modern beekeeping, honey production and treatment of bee diseases (1). With the transition of beekeeping from traditional production to modern production, there has been a rapid increase in the world's bee population. In addition to the increase in the number of bee colonies, disruptions in queen bee production, lack of control

in queen bee sales, buying and selling of bees between countries, increase in migratory beekeeping, development of industry, shrinkage in beekeeping areas, and widespread use of pesticides due to agricultural pesticides have caused an increase in problems in the beekeeping sector. While some of these effects are partially effective, beekeeping has been exposed to another threat with global warming, which has become increasingly important in recent years. Bee diseases have become increasingly important in recent years (2).

While our country did not recognize the varroa parasite until 1976, the parasite has become the most important problem of our beekeeping. A similar situation occurs in many countries. Newly emerging diseases may come to our country after a while. For this reason, veterinarians have important duties. Clinical findings can help in the diagnosis of the disease, but the definitive diagnosis is made by laboratory examinations of the samples sent. The correct diagnosis of any bee disease depends on the sample submitted. Honey bees are social creatures that live in colonies. There are three different individuals in a bee colony. These individuals differ in morphological, physiological and behavioral aspects. Under normal conditions, colonies contain 1 queen bee, tens of thousands of worker bees, the number of which varies depending on the season, and 0-5000 male bees, the number of which varies depending on the season and need. The honey bee life cycle begins as an egg in all individuals. The egg stage is followed by the larva and then the pupa stage. Finally, an adult bee emerges. The queen bee becomes an adult in 16 days, the worker bee in 21 days, and the drone in 24 days. While queen bees can live for 5-7 years, the average lifespan of worker bees is 1-1.5 months. Queen bees and drones weigh an average of 200 mg, while worker bees weigh an average of 120 mg. While queen bees have a long and narrow body structure, the body of male bees is a little more plump. Bee colonies are communities that live with a certain hierarchy and order. Female individuals do the main work in the colony (2).

Male bees have no duties other than mating. The most important duty of the queen bee, one of the colony individuals, is to ensure the continuity of the colony by laying eggs. A high quality and young queen bee can lay an average of 1500-2000 eggs per day. They also maintain order in the hive thanks to the pheromones they secrete. All other work in the hive, including brood feeding, royal jelly and wax secretion, honeycomb cleaning, ventilation, defense, nectar collection-storage-ripening, pollen-propolis collection, water collection,

and even the feeding and care of the queen bee are carried out by worker bees (2).

Although the colony maintenance and controls to be carried out in the hive vary according to seasons and conditions, the following are the points that should be taken into consideration almost every time.

1. **Queen Bee Existence and Productivity;** The most important individual in colony productivity is the queen bee. The first thing to do in in-hive colony maintenance and control is always to check the existence and productivity of the queen bee. The most accurate way to detect the presence of the queen bee in colony controls is to see it face to face. If the queen bee cannot be seen when the hive is opened, we can confirm the presence of the queen bee in the colony by observing whether there are eggs in the honeycombs. However, even if we cannot see the queen bee for some reason, we can make comments about the queen bee from the sound inside the hive, the order in the hive, and the behavior of the bees. in beekeeping
2. The first condition for success is the use of young and high-quality queen bees. The simplest way to measure whether the queen bee is young and of high quality is to observe the eggs she lays in the cells of the honeycomb. If the eggs in the honeycomb are regular and numerous, the queen bee is considered young and of high quality and is kept in the hive. If the eggs in the comb cells are scattered or few, they should be monitored with the assumption that there may be a problem with the queen bee. If this situation continues, the queen bee of the colony will be young and of high quality, placement mistakes should not be made, frames with broods should be kept side by side for brood warming and care, and the brood area within the hive should not be divided. However, if a negative or messy situation occurs in the hive for any reason, the frames in the hive must be rearranged or replaced with new frames if necessary (2).
3. **Honey-Pollen Need of the Colony, Honey and Pollen Storage-Stock Status in Frames;** Bees produce honey, pollen and other bee products not only for humans. Colony individuals in the hive will need these products at various periods. If these substances are not found in the hive at the right time, the development and health of the colony will be in danger. For this reason, the honey and pollen stock stored in the hive and in the comb cells should be observed, and if there is a deficiency, it should be supplemented either from other healthy colonies or in the form of nutritional supplements (2).

4. **Disease and Pest Control and Combat;** One of the most important issues in in-hive controls is disease and pest control. Especially brood diseases and varroa pests must be observed in the comb cells. Even if the varroa pest is not visible to the naked eye or detected in the hive, control must be made at appropriate seasonal times. Care should be taken to use timely and legally permitted drugs against diseases and pests (2).

Clinical findings can help in the diagnosis of the disease, but the definitive diagnosis is made by laboratory examinations of the samples sent. The correct diagnosis of any bee disease depends on the sample sent (3,4).

Where the morbid substance was taken from, the name of the bee business or owner, the number of hives, the date the morbid substance was taken, the symptoms of the disease, which diseases are suspected, whether the disease is a larval or adult bee disease, whether there is a similar disease or death in other bee colonies in the vicinity, the collective In cases of death, it should be stated in detail in the morbid substance protocol whether agricultural pesticides have been applied in the environment or not, and whether there is sufficient food (pollen and honey) in the hive in cases of deaths in the winter season (Table 1), (3,4).

1. **Adult Bee:** In case of adult bee diseases, at least 50-60 adult bees from sick and dead bees should be sent to the laboratory in a small cardboard box for each suspected disease (3,4).
2. **Comb with Young:** In cases of disease and death in larvae, the comb containing dead or diseased larvae should be sent to the laboratory as a whole, if possible, or in half a frame, if possible, and sent to the laboratory in a wooden or cardboard box. Care should be taken to ensure that the honeycombs contain as little or no honey as possible. For virological examination, the samples to be sent must be sent from at least 50 live bees, without adding any substance, and in compliance with the cold chain conditions (Table 1), (3,4).

Table 1. Örnek Protokol (3,4).

LABORATORY MEDICAL SUBSTANCE FOR BEE DISEASES SENDING PROTOCOL	
Name and Surname of the Beekeeper	
Beekeeper's Address	
Number of Colonies in the Apiary	
Origin of Bees if Mobile Beekeeping is Practiced	
Onset Date of the Disease	
Number of Sick Colonies in the Apiary	
How Many Colonies Failed Due to Disease?	
Medicines Used and Expiration Dates	
Clinical Findings	
Which Disease Is Suspected?	
Type and Number of Diseased Substance	
Date of Receipt of the Morbid Substance	
Thoughts of the Veterinarian Who Received the Morbid Substance:	Veterinarian Signature

Suspicious materials collected for diagnostic purposes should be forwarded to a nearby diagnostic laboratory as soon as possible (4).

Important criteria to consider when purchasing materials are;

- Material appropriate to the pathogenesis and clinical course of the suspected infection should be collected.
- The instruments to be used in collecting suspicious material (scissors, scalpel, syringe, stool container, urine container, swab, etc.) must be sterile.
- The date and time of material collection and the region from which the material was taken must be written on the container from which the material is taken, and the container must be tightly closed and taped.
- In the document to be sent with the material received, the disease suspected should be stated, the anamnesis of the disease and the patient's description (age, gender, address) should be written. The physician receiving the material must sign the document (4).

Processing of adult bee and brood comb materials in the virology laboratory

Adult bee or honeycomb materials used in inoculum preparation must be fresh. After the adult bee or honeycomb materials are taken to the virology laboratory, they are processed by following the steps below (Figure 1) (4).

- a. All tools and equipment to be used in processing the materials must be sterile.
- b. The outer surface of adult bees or honeycomb materials intended to prepare inoculum is disinfected by wiping with alcohol.
- c. Approximately 1 gram of honeycomb or 10-15 dead adult bee samples are taken with sterile forceps and scalpel and read into PBS containing concentrated antibiotics.
- d. It is crushed thoroughly using a mortar or homogenizer. If the process is carried out in a mortar, sterile sand must be added to the mortar. This will allow the organ part to be crushed more easily to after the chopping and crushing process is completed, it is filtered into sterile containers with cheesecloth stretched over the mouth. Thus, the coarse particles will remain in the cheesecloth and the liquid part will accumulate in the container. This liquid is then transferred into tubes.
- f. It is centrifuged at 3000 rpm for 30 minutes at 4°C.
- g. At the end of centrifugation, the upper liquid part is removed. 10% concentrated antibiotic is added into it. It is kept at 4° for 1 hour for the antibiotic to take effect.
- h. Then, this inoculum is inoculated into agar to check whether there is any bacterial contamination. Suspicious organ samples with no bacterial growth on the agar after 24-48 hours are ready for isolation. Stored at -80° until work is done. Inoculums with bacterial contamination are filtered in filter systems that bacteria cannot pass through for sterility (4).

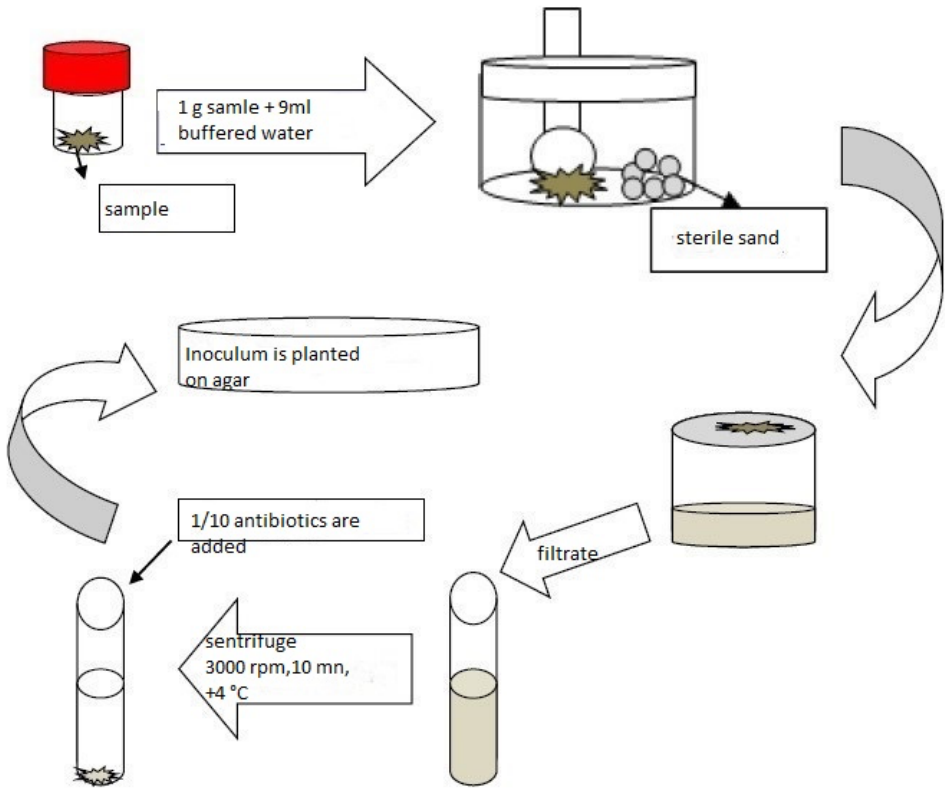


Figure 1: Preparation of inoculum from honey bee or honeycomb sample (4).

Laboratory Diagnosis

Determining the nucleic acids of honey bee viruses using molecular methods and performing sequence analysis (3,5).

Diseases in honey bees occur due to many different factors. The virus can be found in bees without causing clinical symptoms. These viruses are transmitted vertically over several generations in bees. While the disease is latent in this way, as the number of the virus increases, the disease ceases to be latent and develops clinical symptoms. The emergence of these clinical symptoms occurs due to environmental factors affecting the bee. Therefore, causes such as environment, colony, pathogens are only a link in the chain in the emergence of the disease. For this reason, the investigation of the cause of bee diseases should be examined as a whole. (figure 2). The location and environment of the colony settlement, food diversity, weather conditions, density of beehives in the region,

accessible water, climate and wind, hive model, material and general condition of the hive are very important about the health of the colony. In the examination of colony viral diseases, anamnesis, examination of the first pathological findings, examination of the hive entrance hole, the number of bees entering and exiting, the way the bees fly, provide an impression about the strength of a colony. Unusual flying patterns may be a neurological sign of viral diseases. The waste in front of the hive should be investigated for dead bees and larval mummies. These observations provide important information about colony health. When examining the honeycombs, the color of the wax is dark due to its use in several incubation periods. By evaluating pollen and honey stores, an idea about the nutrition of the colony can be obtained (Figure 2),(3,5).

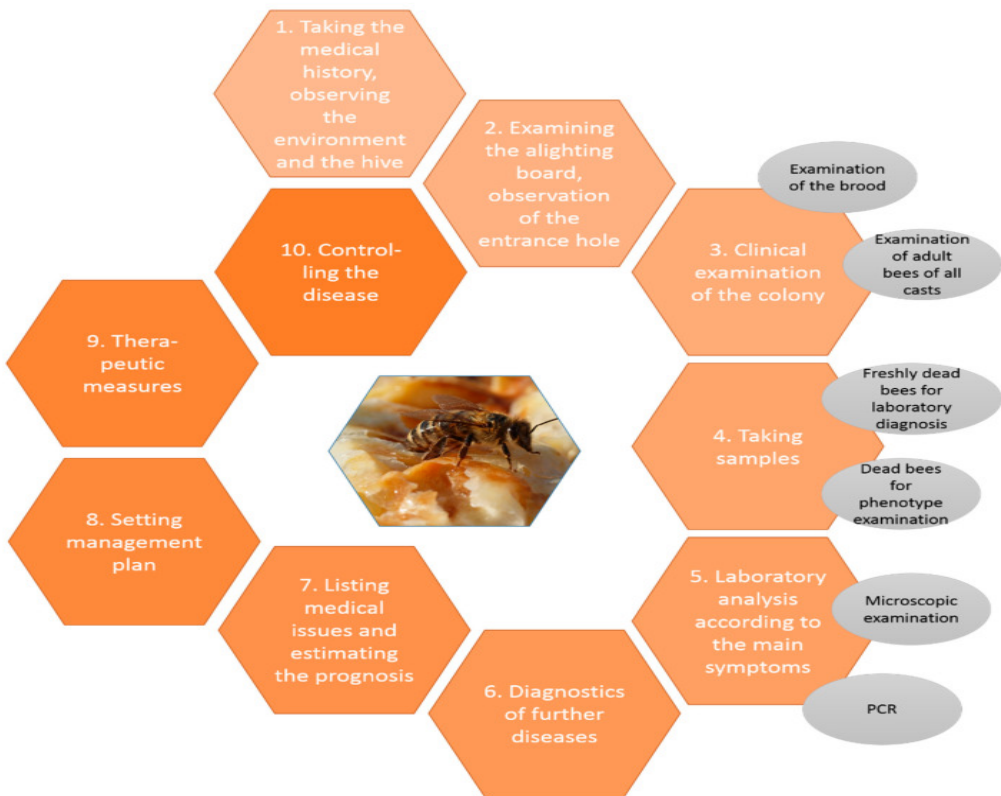


Figure 2: *Diagnostic approach in bee diseases. (5).*

PCR to Detect Bee Viral Diseases

Deoxyribonucleic acid (DNA) consists of two parts: base pairs and a sugar-phosphate backbone. The bases in the DNA molecule are pyrimidine (Cytosine and Thymine) and purine (Adenine and Guanine) bases. In normal DNA structure, the Guanine (G) - Cytosine (C) pair is approximately the same size as the Adenine (A) - Thymine (T) pair, and G always pairs with C and A with T. The structure in which a pyrimidine or purine base is combined with a pentose sugar is called a “nucleoside”. The compound formed by combining nucleosides with phosphate molecules is called “nucleotide”. Nucleotides located next to each other form polynucleotide chains by combining with covalent phosphodiester bonds extending from the 5'carbon end of one nucleotide to the 3'carbon end of the other nucleotide. A DNA molecule consists of two antiparallel polynucleotide chains that are helically linked and complementary to each other. A typical animal cell has a DNA consisting of approximately 3×10^9 nucleotides. The DNA molecule has approximately 10 nucleotide bases in each helical turn and is 3.4 mm long^{1,2}. In the DNA molecule, a spiral strand is connected to the other strand by non-covalent hydrogen bonds between opposite bases (A and T are connected to each other with two hydrogen bonds, G and C with three hydrogen bonds). Hydrogen bonds, which form as a result of hydrogen moving towards positive current when bonded to electro-negative atoms such as oxygen and nitrogen, have approximately 1/10 the strength of the covalent phosphodiester bonds of the sugar phosphate backbone. This feature allows the strands to be separated from each other without damaging a single strand in the double helix structure of DNA (6).

PCR, one of the DNA-based tests developed by taking advantage of the basic physico-chemical behavior of DNA, was reportedly first applied by Kary Mullis in 1985. PCR is a nucleic acid technology method used to amplify gene segments between two known DNA sequences. In the field of virology, in vitro enzymatic amplification of DNAs or some specific gene segments belonging to factors found or isolated in clinical materials is performed by PCR. This technique can increase the amount of template DNA at the nanogram level in the total genomic DNA by 10⁵-10⁶ times. The first information to know in the technique is what the target DNA sequence is. Thus, the primers (precursor DNA molecules) required for the reaction are synthesized using a DNA synthesizer (6,7,8).

Almost all bee viruses carry a positive-strand RNA genome. More than 20 bee viruses have been identified that cause disease in bees. Acute bee paralysis virus (ABPV), Kashmir bee virus (KBV), Israeli acute paralysis virus (IAPV), forming the ABPV-KBV-IAPV complex, Black queen cell virus (BQCV), Chronic bee paralysis virus (CBPV), Deformed wing virus (DWV) and Sac brood virus (SBV) are some of the important ones in terms of causing disease. This book chapter will focus on diagnostic methods of ABPV-KBV-IAPV, CBPV and DWV in veterinary medicine (Table 2).

Table 2. Overview of selected honey bee viruses (taxonomy, symptoms, affected bee species and modes of transmission), (3, 12).

Virus	Taxonomy	Symptoms in a Colony	Affected Castes	Transmission (Main Routes)
Chronic bee paralysis virus (CBPV)	unclassified (RNA)	hairless black syndrome and paralysis syndrome, high mortality after a few days	w, d, q	c, o
Acute bee paralysis virus (ABPV)	<i>Dicistroviridae</i> (RNA)	paralysis and high mortality after 1–2 days (experimentally)	w, d	vec, ver (v, to)
Kashmir bee virus (KBV)	<i>Dicistroviridae</i> (RNA)	mortality without other symptoms	w, d	vec, ver
Israeli acute paralysis virus (IAPV)	<i>Dicistroviridae</i> (RNA)	paralysis and death	w, d	vec, ver
Deformed Wing Virus (DWV)	Picornia-like virus (RNA)	crippled bees with deformed wings and shortened abdomens	w, d, q	vec, ver (v, to)

[Open in a separate window](#)

Legend: RNA = Ribonucleic acid, w = worker, d = drone, q = queen, c = direct contact, o = oral-fecal, vec = vector-borne, ver = vertical, v = venereal, to = transovarial.

The causative agents of viral diseases of honeybees have an RNA genome. In order to perform PCR in the diagnosis of these factors, total RNA is purified from adult bees or larvae. PCR test is applied after the RNA is converted into cDNA with the Reverse Transcriptase enzyme. For this reason, the test is called RT-PCR (13).

Table 3. Specific primers of bee viruses (9,10).

Primers and target genes	Sequences	tm
CBPV (RdRP)	F: GCAAACGCCCACCAATAGT R: TGGTACGGAAGGTGTGTCAA	55°C
SBV (cp gene)	F: TATTCAGGGGGACGCTACAC R: AGTGCTGCTTGAAACCTGT	
IAPV (cp gene)	F: TTGGCGTGCAACTATGTGTT R: TCTTCTGCCCCTTCCAAAC	
BQCV (cp gene)	F: GACAGCGTGCCAAAGAGAG R: GCGAACCCGTCCAATACTTA	
KBV (cp gene)	F: CACATTCCGAACAATAA R: GCGATAGGAATTTTGCGGTA	
DWV (Non-structural protein)	F: TTGGTATGCTCCGTTGACTG R: ATTCCTCAGAAGTTGGTTTCG	
ABPV (cp gene)	F: GTATGGAAGTGGGCTGAGGA R: CGCGGTACTAAAAAGCTACGA	

After total RNA isolation, cDNA is obtained and the genome of the virus is amplified by PCR using specific primers (Table 3).

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Chapter

5

BACTERIAL DISEASES OF HONEYBEES AND BIOCHEMICAL TESTS THAT PLAY A ROLE IN THE DIAGNOSIS OF DISEASES

VeySEL TAHİROĞLU

Introduction

Beekeeping is an agricultural activity carried out in almost every region of the world (1). Beekeeping activity dates back to ancient times in world history. According to studies; The relationship between bees and humans dates back to B.C. It is reported that it dates back to 7000 years ago (2,3). Like all living things, honeybees (*Apis mellifera*) also face disease risks and epidemic threats (4). Unlike other animals, honey bees are considered a super-organism consisting of thousands of individuals and include queen, worker and drone bees (5). Due to the density of bees in the hive, their close settlement and their foraging habits, infections spread easily within and between hives. Adult bees that exhibit social immunity-related behaviors and cooperate in infected hives cause bacterial diseases to be ignored (6). Bacterial diseases that have medical importance in

honey bee colony health are generally considered as brood diseases. Among these, American foulbrood and European foulbrood diseases have serious economic and medical importance. There are clinical similarities between the symptoms that are important in the control of both diseases. In some countries such as Canada, Australia, England and the United States, the use of antibiotics is allowed in colonies for pollination and package beekeeping purposes. In countries like Türkiye, where beekeeping is directly related to human consumption, the use of antibiotics in the fight against diseases has been banned for about thirty years. Therefore, the importance of biochemical parameters, molecular tests and microbiological tests is increasing day by day to quickly and accurately diagnose bacterial diseases. In this section, information will be given about important bacterial diseases of honey bees and their diagnostic methods.

1. BACTERIAL DISEASES OF HONEYBEES

1.1. American Foulbrood

This disease is a very dangerous and epidemic brood disease that occurs in honey bee larvae and causes the larvae to die and stink. All three developmental periods are affected by the disease. The cause of the disease is a spore-forming bacterium called *Paenibacillus larvae*. It was first discovered in 1906 by American Dr. G.F. White (7). *Paenibacillus larvae* is a brood pathogen for the entire *Apis* genus and is inactivated in the normal intestinal flora of healthy adult bees. It can be neutralized outdoors at high pressure and temperature (121°C, 1.5 atm, 15 min) or by applying suitable chemicals or UV light. Logarithmically, 250 million new bacilli can develop from a spore in an infected bee larva within 24 hours. While young larvae can be infected with a small number of spores, old larvae can only be infected with millions of spores. The onset of disease symptoms is a separate issue, and 50 million spores are generally sufficient, the amount of which varies depending on the strain of the agent. The more spores the larva is initially infected with, the shorter the logarithmic reproduction rate and the time it takes for symptoms to appear due to the strain. 3 billion spores are produced even in the larva, which dies in varying periods of time depending on the strain of the agent, but within 6-7 days at the earliest. Colonies with hygienic behavior can transmit billions of spores to the beeswax, honey, the bodies of other bees, and the offspring and larvae they feed, during the cleaning of dead

larvae and the transportation of rotting offspring from the hive to the outside. Infection can occur even after 30 years in a hive contaminated with spores, and the presence of spores in soil and similar environments varies depending on external factors. This is enough to understand why contaminated hives and contact materials should be burned (8-12). The primary clinical finding for diagnosis is that the infected larva usually takes on a brownish, semi-fluid glue-like consistency after death. The honeycomb eyes of infected colonies take on a patchy appearance. A dark-greasy appearance and inward depression can be seen in the closed brood eyes containing diseased honey bee larvae. When a matchstick is inserted into the brood eyes and pulled out, creamy or dark brown glue-like larval residues that grow in the form of threads can be easily diagnosed by beekeepers as the most obvious clinical finding. It is reported that when the larva decays during the pupal period, it characteristically takes on the appearance of a pupal tongue extending to the top of the brood (8,13). It is reported that the first official record of American foulbrood in Turkey belongs to the diseased honeycomb sample sent from the Pınarhisar district of Kırklareli in 1947 (14).

1.2. European Foulbrood

European Foulbrood is a bacterial disease that affects honeybee larvae. European Foulbrood, simply called EFB, is a bee disease that causes huge economic losses. It has spread over a wide area all over the world. The disease was first described by Cheshire and Cheyne in 1885. However, at that time, the causative agent was thought to be *Bacillus alvei*, one of the many bacteria associated with European foulbrood. In 1956, the main disease agent was identified and identified as *Bacillus pluton*. In the same year, Bailey renamed the bacterium *Streptococcus pluton* because of its cocci-like shape. The bacterium that causes EFB was last reclassified by Bailey and Colins in 1982 and named *Melissococcus plutonius*. The name European foulbrood was used by Philips because Europe was the place where the disease was first discovered (15,16). It is gram-positive, usually spindle-shaped and sometimes pleomorphic and rod-like in appearance. They appear under the microscope individually, in pairs, or in chains of various lengths. The optimum growth temperature is 35°C, it is a microaerophilic bacterium that can also grow anaerobically, and it needs carbon dioxide to grow. The disease is usually seen in spring and the first half of summer, when the

nectar flow begins, when brood rearing is intense. The disease occurs in nectar shortage and cold weather conditions. When the amount of nectar coming into the hive reaches the highest level, the severity of the disease decreases. Since diseased larvae are often removed from the hive by bees before any symptoms of the disease appear, the disease may not be noticed. Bee larvae take the agent into their digestive system through the food carried by the bees during feeding. The agent, which settles in the digestive system of the larva, develops in the intestine and is excreted into the comb chamber with feces after the fry enters the pupal period. While worker bees clean these residues from the honeycomb cells, they transmit the disease to healthy larvae. Adult bees that are carriers are not affected by this disease. The disease affects 3-4 day old larvae more. Larvae are usually infected with the disease when they are 3-4 days old. The areas with brood in the comb are not regular. Patchy offspring pattern is seen. Old larvae, young larvae, eggs and empty cells are together in the same place in the honeycomb. 90% of pup deaths occur in open pup eyes. However, if the disease is severe and advanced, deaths can also occur in closed puppy eyes. If death occurs in the coated puppy's eyes, the glaze layer of the eyes collapses and their color becomes lighter. Closed puppy eyes become pierced due to secondary factors. Dead larvae lose their bright pearl white color and their bodies turn transparent. Their intestines and tracheas are visible from the outside (17-19).

1.3. Septicemia

Septicemia is an adult honey bee disease caused by bacteria called *Pseudomonas aeruginosa* (= *Pseudomonas apiseptica*). *Pseudomonas apiseptica* is a gram (-) and non-spore-forming bacteria. This bacterium is found in nature in moist soils, plants, stagnant water and swamps. *Pseudomonas apiseptica* causes disease by passing into the respiratory (tracheal) system of the bee through various means and from there into the blood fluid. The disease is seen in hives that are airless and have high levels of humidity. In addition, stress factors such as intense artificial feeding, adverse weather conditions, honeycomb weaving stress and the onset of varroa damage increase susceptibility to septicemia. The disease is seen in all three development stages of the bee. Bees infected with the disease die in a short time. Deaths mostly occur within 20-36 hours after infection. While the blood color in healthy bees is pale yellowish or amber, in sick bees the

blood color changes from brown to chalk white. The most important symptom of the disease is the degeneration of the muscles. That's why it's impossible to handle dead bees. When held by hand, the legs, wings, head, chest and abdomen of the bees are immediately separated (15,20,21). *Serratia marcescens*, another causative agent of septicemia, is a Gram-negative opportunistic pathogen of a variety of animals, including humans and insects. In most animals, *S. marcescens* is lethal only when present in the bloodstream. The *Serratia* genus are rods that generally produce prodigiosin, the characteristic red or pink pigment. *S. marcescens* has also been isolated from diseased honey bee larvae. *S. marcescens*, which, along with Enterobacteriaceae, is considered an indicator of atypical microbiome composition in bees, is commonly found at low frequencies in honey bee intestines. It has been reported that since the *S. marcescens* kz11 strain isolated from honey bees shows high resistance to tetracycline, it causes increased mortality in bees treated with tetracycline (22-26).

1.4. Spiroplasmosis

Spiroplasma apis and *Spiroplasma melliferum* are honey bee pathogens. Spiroplasmas are small, spiral, motile, Gram-positive bacteria that lack a cell wall. They multiply by passing through the intestinal barrier and reach the hemolymph, causing the death of the bee. *S. melliferum* was first reported in honey bees in 1976 during an examination of honey bees for pathogenic microorganisms. *S. apis* was detected abundantly in honey bees in colonies showing "May disease" symptoms in France in the early 1980s. This disease is also called "May disease" because it occurs in May and June. *S. apis* B31 strain was found to cause death in honey bees when injected, and B39 strain when fed. *S. melliferum* caused similar disease symptoms in bees when fed, but was less pathogenic at the colony level. However, available information on the pathogenesis of spiroplasmas in infected honey bee colonies is limited, and no significant bee colony losses associated with spiroplasma infection have been observed (27 – 30).

2. DIAGNOSIS OF BACTERIAL DISEASES OF HONEYBEES

2.1. Symptoms and Biochemical Tests of American Foulbrood Disease

Bacterium	<i>Paenibacillus larvae</i>
Clinical Findings	❖ The primary clinical finding for diagnosis is that the infected larva usually takes on a brownish, semi-fluid glue-like consistency after death. The honeycomb eyes of infected colonies take on a patchy appearance. A dark-oily appearance and inward depression may be observed in the closed brood cells containing diseased honey bee larvae. Creamy or dark brown glue-like larval residues that grow into threads when a matchstick is inserted into the young eyes and pulled out can be easily diagnosed by beekeepers as the most obvious clinical finding. It is reported that when the larva decays during the pupal period, it may characteristically appear as a pupal tongue extending to the top of the brood (8,13).
Biochemical Tests and Molecular Diagnosis	❖ Although it varies between <i>P. larvae</i> genotypes, they have a typical carbohydrate fermentation profile with acid production from glucose and trehalose, but they cannot produce acid from arabinose and xylose. In addition, catalase is negative and the factors can hydrolyze casein or milk. The Holst milk test is one of the simple laboratory tests based on the detection of proteolytic enzymes that hydrolyze casein in the sporulation of <i>P. larvae</i>. Approximately 5% milk solution is prepared, poured into two glass tubes and diseased larvae are added to one. It is considered positive if the liquid in which the larvae are placed turns clear brown after waiting for 1 hour (31). ❖ The first PCR test for the identification of larvae was performed by Govan et al. (1999) and is based on the 16S rRNA gene(32). ❖ Erban et al. (2017) evaluated the relative proportions of <i>P. larvae</i> in the total body microbiota of honey bees using 16S rRNA metabarcoding analysis (33). Beims et al. (2020) in their study, they developed multiplex qPCR, a new multiplex diagnostic method, to distinguish <i>P. larvae</i> genotypes ERIC I and ERIC II (34). Kušar et al. TaqMan-based qPCR developed by (2021) was calibrated using digital PCR (35). “The Vita American Foulbrood Diagnostic Test Kit” has been developed commercially for the diagnosis of the disease. The suspicious larval sample is placed in the extraction bottle and shaken for 20 seconds. 2 drops of the bacterial suspension are placed on the test apparatus held horizontally, and the double blue line seen after 1-3 minutes indicates that it is positive (31). ❖ Pastucha et al. (2021) prepared a specific antibody for <i>P. larvae</i> and used it to develop an upconversion-linked immunosorbent assay (ULISA). For this purpose, Photon-upconversion nanoparticles (UCNP) were conjugated to streptavidin via a PEG-linker using copper-catalyzed click chemistry. It is reported that this developed method showed low cross-reactivity and provided a detection limit of 2.9×10^3 CFU (Colony-forming unit)/mL, which is 22 times more sensitive than ELISA (36).

2.2.Symptoms and Biochemical Tests of European Foulbrood Disease

Bacterium	❖ <i>Melissococcus plutonius</i>
Clinical Findings	❖ Diagnosis is based on visual inspection of the comb and detection of diseased larvae. The general symptoms in the colony are the irregular distribution of lidded and lidless eyes in the brood. Materials such as a 10x10 cm piece cut from the area where symptomatic baby eyes are located, nurse bees responsible for the baby eyes, honey and pollen can be used for diagnosis (37). ❖ Young larvae that die from infection cover the bottom of the cell and are almost transparent enough to see the trachea (Forsgren et al., 2013). Older larvae, on the other hand, die by assuming a bent or stretched shape around the walls, instead of the normal spiral position of the larva in the hatchling's eyes. The color of the larva changes from pearl white to yellow, then brown and finally black (31,37).
Biochemical Tests and Molecular Diagnosis	❖ Molecular methods such as classical PCR (Govan et al., 1998), hemi-nested PCR (Djordjevic et al., 1998), and real-time PCR (Roetschi et al., 2008) can be used for diagnosis (38-40). ❖ Arai et al. Duplex PCR developed by (2014) provides the distinction between typical and atypical strains (41). ❖ Milbrath et al. (2021) compared the lateral flow device (LFD) developed for field diagnosis with microscopic examination and qPCR in field samples. They found a high level of agreement between these three diagnostic techniques (42). ❖ As an immunology-based method, Tomkies et al. (2009) optimized the Lateral Flow Device (LFD), which detects biological antigens in infected tissue extracts, for field diagnosis of the disease by producing a monoclonal antibody specific for <i>M. plutonius</i>. Thus, a new diagnostic method, LFD, was developed with these optimized antibodies (43). ❖ Although ELISA is one of the best methods used for antibody-based bacterial detection, it is not sensitive enough to reveal early infection. In order to overcome this deficiency in the ELISA method, Poláchová et al. (2019) have developed a method in which photon-upconversion nanoparticles (UCNPs) can be used as ultrasensitive labels in immunoassays after conjugation with specific biological recognition molecules. Researchers have reported that this method, called ULISA, is 400 times more sensitive than the traditional ELISA method and has great potential for both early diagnosis of the disease and preventing its spread (44).

2.3. Symptoms of Septicemia and Biochemical Tests

Bacterium	❖ <i>Serratia marcescens</i> ❖ <i>Pseudomonas apiseptica</i>= <i>Pseudomonas aeruginosa</i>
Clinical Findings	❖ The symptoms of the <i>Serratia marcescens</i> agent have been observed to be a decrease in the movements of worker bees and drones and their separation from other active hive members. In December and January, symptomatic workers were found motionless on top of the inner cover, although they remained alive. Additionally, male bees showing symptoms were unable to fly during the warmer months. Despite the infection, it was not observed that infected bees were forcibly removed by guards and workers responsible for cleaning the hive. However, freeing themselves and leaving the hive supports hygienic behavior that is thought to limit disease transmission (45). ❖ <i>Pseudomonas aeruginosa</i> The most important symptom of the disease is degeneration of the muscles. That's why it's impossible to handle dead bees. When held by hand, the legs, wings, head, chest and abdomen of the bees are immediately separated (15).
Biochemical Tests and Molecular Diagnosis	❖ Burritt et al. (2016) identified a new strain, <i>S. marcescens</i> sicaria strain (Ss1), by interpreting the 16S rRNA sequence and whole genome sequence analysis of a Gram-negative bacterium isolated from the hemolymph of worker bees. This new strain showed several phenotypic and genotypic differences, including 65 genes not previously found in <i>Serratia</i> genomes. <i>S. marcescens</i> Ss1 strain was isolated especially from Varroa mites in wintering hives and from the hemolymph of inactive or dead bees (45). ❖ Raymann et al. (2018) observed that <i>S. marcescens</i> Ss1 exhibited virulence only when found in the hemolymph of honey bees. In this case, they reported that they concluded that it had different virulence and infection mechanisms than intestinal isolates (26).

2.4. Symptoms and Biochemical Tests of Spiroplasma Disease

Bacterium	❖ <i>Spiroplasma apis</i> ❖ <i>Spiroplasma melliferum</i>
Clinical Findings	❖ In <i>S. apis</i> infection, in which worker bees are affected, there are many dying or dead bees in the hive, and it has been observed that infected bees cannot fly but can only vibrate in front of the hive (29). ❖ Their abdomens are hard and swollen, and their intestines are filled with undigested pollen. It is reported that while individual bees die, the colony recovers spontaneously over time (31).
Biochemical Tests and Molecular Diagnosis	❖ Meeus et al. (2012) designed a multiplex PCR based on species-specific primers to amplify the <i>S. melliferum</i> spiralin and <i>S. apis</i> rpoB genes, and these were combined with universal primers against <i>Spiroplasma</i> spp., allowing the detection of both species after the isolation of the bacteria. This method can identify <i>spiroplasma</i> from cultures without the need for DNA extraction and in as little as 3 hours. Thus, direct culturing of hemolymph is lower cost since the DNA extraction step is eliminated (46). ❖ Schwarz et al. (2014) enabled species-specific high sensitivity detection with a qPCR test using <i>S. melliferum</i> spiralin primers and <i>S. apis</i> 16S rRNA primers (47).

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BACTERIAL DISEASES AND TREATMENT METHODS IN HONEYBEES

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Introduction

Honeybees are an indispensable part of natural ecosystems and play a crucial role in agricultural production. They assist in maintaining plant diversity through pollination and contribute to the efficient growth of many agricultural crops. However, in recent years, a range of factors threatening the health of honeybees has emerged. Among these threats, the spread of bacterial diseases can lead to the weakening and even death of bee colonies. This situation is a significant concern for the beekeeping industry, necessitating the development of effective treatment methods to protect bee health(1, 2, 3).

In this introductory article, we will discuss bacterial diseases observed in honeybees and the treatment methods used to combat these diseases. To achieve this, we will first identify commonly observed bacterial diseases in honeybees and discuss their effects on bee colonies. Subsequently, we will examine various treatment methods used in combating these diseases and provide information on their effectiveness. Finally, by discussing future research and application

areas, we will make recommendations for developing more effective strategies to preserve bee health. This introductory article will provide a foundation for understanding the significance of bacterial diseases in honeybees and the treatment methods used to combat them(4, 5, 6).

Healthy honey bee colonies hold ecological and economic value not only in increasing animal and agricultural production efficiency but also in maintaining plant diversity. Bacterial diseases of medical significance in honey bee colony health are generally considered brood diseases. Among these, American Foulbrood (AFB) and European Foulbrood (EFB) are of significant economic and medical importance(56, 57, 58). There are clinical similarities in symptoms crucial for controlling both diseases. In some countries, such as Canada, Australia, the United Kingdom, and the United States, the use of antibiotics in colonies for pollination and package beekeeping purposes is permitted. In countries like Turkey, where beekeeping is directly related to human consumption, the use of antibiotics against diseases has been banned for approximately twenty-five years(65, 64, 59). Therefore, colony-level preventive medicine, good beekeeping practices, and hygiene applications are extremely important. Worldwide, the diagnosis and treatment of animal diseases fall under the authority of veterinarians. In Turkey, official diagnoses are made by Veterinary Services units, which provide reference laboratory services for notifiable diseases(60, 63). The terms “American” and “European,” found in historical records from about a century ago, were used not to denote the origins of the pathogens but to indicate where they were identified and studied(61, 62). These diseases encompass infectious diseases that are significant in international animal and animal product trade and are considered risky in terms of socio-economic and public health. At this point, the health of bee colonies must be sustainably maintained (77, 76, 75, 74). In Turkey, European Union and OIE standard regulations are applied in the fight against AFB, which is a notifiable disease. It is a current necessity for veterinarian colleagues to be well-informed about the aforementioned notifiable diseases that fall within the scope of Veterinary Medicine for Honey Bees (VMHB) in terms of legal authority(73, 72).

A bee hive is considered a superorganism composed of thousands of individuals, including developing larvae and adult bees, rather than individual animals treated separately by veterinarians. In this superorganism, which includes the queen, worker, and drone bees, larvae are generally the most concerning

and need treatment (46, 71). The primary agents identified as causing bacterial diseases in honey bees are *Paenibacillus larvae*, *Melissococcus plutonius*, *Serratia marcescens*, *Spiroplasma apis*, and *Spiroplasma melliferum* (47, 70). Among these, the most significant in terms of economic losses due to the diseases they cause are *P. larvae* and *M. plutonius*, which mostly affect larvae (48, 69). Infections spread easily within and between hives due to the density of bees, their close proximity, and their foraging habits. Adult bees exhibiting behaviors related to social immunity and cooperation in infected hives often lead to the neglect of bacterial diseases. When exposed to pathogens, adult worker bees control the hive entrance to attack or exclude infected bees (49, 68). This behavior makes it difficult to detect bacterial pathogens in infected colonies (51). Additionally, different pathogens often infect a bee hive simultaneously, further compromising colony health and making it more vulnerable to other challenges (50). Due to the importance of bee products and their role in pollination, beekeeping is a significant sector globally. The international trade of honey bees leads to the global spread of infections, many of which cause substantial losses for beekeeping (52, 53, 67). Therefore, diseases that cause productivity losses in honey bees are of serious importance. Consequently, accurate diagnosis of the most important of these diseases is necessary. This long-standing necessity has encouraged researchers working on bee diseases to develop fast, practical, and highly reliable methods and tools. This review focuses on diagnostic methods and current developments for bacterial bee diseases(54, 55, 66).

Bacterial Diseases in Honeybees

The diversity of bacterial diseases observed in honeybees is quite extensive. Among these, the most common ones include American Foulbrood, European Foulbrood, and Acute Paralysis Virus. These bacterial diseases can cause significant damage to bee colonies and even colony losses(7, 8, 9).

American Foulbrood is an infectious disease caused by the bacterium *Paenibacillus larvae*. It is a bacterial infection that leads to the death of bee larvae. European Foulbrood, on the other hand, is a disease caused by the bacterium *Melissococcus plutonius* and is often associated with deficiencies in bee larval nutrition. Acute Paralysis Virus is a viral disease that weakens the immune system of bees and can lead to the death of colonies(13, 14, 15).

Treatment Methods

Various methods are used for treating bacterial diseases in honeybees. Antibiotics can be applied to colonies showing signs of disease. For example, antibiotics such as oxytetracycline and tylosin can be effective in controlling American Foulbrood. However, it should be noted that antibiotic use may increase the risk of antibiotic resistance development and adversely affect the quality of honey products. Additionally, hygienic practices and colony management play a significant role in preventing the spread of diseases. For instance, isolating colonies showing signs of disease and cleaning infected materials can help prevent the spread of diseases(10, 11, 12).

Future Directions and Recommendations

There is a need for further research to develop effective treatment methods for bacterial diseases. In particular, studies on the effectiveness of natural compounds and probiotics against diseases need to be increased. This can prevent the development of antibiotic resistance and provide safer and more sustainable treatment options for bee health. Furthermore, the importance of hygienic practices and colony management in the beekeeping industry should be emphasized. Beekeepers should regularly inspect their colonies and minimize contact with infected colonies to prevent the spread of diseases(18, 19, 20).

Some areas that future research in combating bacterial diseases should focus on include:

Research on Natural Treatment Methods: Due to concerns such as the development of antibiotic resistance and the passage of antibiotics into honey products, the use of natural compounds and probiotics in the treatment of bacterial diseases should be further researched. The effectiveness of natural products such as plant extracts, propolis, and royal jelly against diseases should be evaluated(16, 17).

Studies on Disease Resistance: Some bee breeds are known to have natural resistance to diseases. Therefore, genetic selection studies should be conducted to increase the disease resistance of bee breeds. Breeding resistant bee breeds can be an effective strategy in combating diseases(29, 30).

Education and Awareness: Beekeepers and beekeepers need to be trained in the recognition and control of diseases. Awareness programs and educational materials should be prepared to increase beekeepers' knowledge of diseases. In this way, early diagnosis and prevention of diseases can be ensured(21, 22).

Applied Research: In addition to laboratory studies, applied research based on field experiments should be conducted. These studies can evaluate the effectiveness of treatment methods under real-world conditions and develop strategies suitable for beekeepers' use. These recommendations highlight areas of focus for future research and efforts to develop more effective strategies for combating bacterial diseases in honeybees. This way, a stronger foundation can be laid for preserving bee health and making the beekeeping industry sustainable(28, 27, 26, 25).

Colony Management and Hygiene Practices Improvement: Beekeepers can prevent the spread of diseases by applying colony management and hygiene practices more effectively. In this context, beekeepers should be educated on colony control and recognizing disease symptoms. Additionally, the selection of colony settlement locations should be carefully evaluated to reduce the risk of disease transmission(23, 24, 31).

Establishment of Disease Tracking and Monitoring Systems: An effective system should be established for tracking and monitoring bacterial diseases in honeybees. This system can enable beekeepers to recognize and intervene in disease outbreaks more quickly. Furthermore, collecting and analyzing disease data is important for determining disease spread trends and predicting future outbreaks(40, 32, 39, 33).

Strengthening Farmer and Beekeeper Collaboration: Honeybees play a significant role in agricultural ecosystems. Therefore, strengthening collaboration and communication between beekeepers and farmers is important. Farmers should understand the effects of agricultural chemicals and fertilizers on bee health and work in collaboration with beekeepers. This collaboration is critical for protecting bee health and ensuring the sustainability of agriculture(34, 35, 36).

Conclusion

Bacterial diseases observed in honeybees pose a serious threat to the beekeeping industry. These diseases can cause significant damage to bee colonies and even colony losses. Therefore, it is important to effectively combat bacterial diseases(37, 38).

In addition to antibiotics, the use of natural compounds and probiotics is important in treating bacterial diseases. However, it should be remembered that the overuse of antibiotics may lead to the development of antibiotic resistance. Therefore, it is important to develop safer and more sustainable treatment options(41, 42).

Furthermore, the importance of hygienic practices and colony management in preventing the spread of diseases should not be overlooked. Beekeepers should regularly inspect their colonies and isolate colonies showing signs of disease(43, 44).

In conclusion, there is a need for further research and implementation to combat bacterial diseases in honeybees. However, the development of effective treatment methods and the widespread adoption of hygienic practices are critical for preserving bee health and making the beekeeping industry sustainable(45, 37).

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VARROA INFECTION DETECTION IN HONEYBEE COLONIES AND CHEMICAL CONTROL METHODS AGAINST VARROA

Münire TURHAN

Introduction

Varroa, whose only breeding and reproduction place is the honey bee colony, multiplies rapidly if the necessary precautions are not taken, causing either the productivity of the colonies to decrease and their extinction in advanced stages, or weakening the colony individuals and making them more susceptible to other diseases. Since these situations are known, many control methods have been developed against the parasite, and chemical control methods are the most popular among them. Chemicals are effective in the fight against varroa, but if the same drug is used several times, the parasite becomes immune to that drug and the drug must be changed.

Unconscious use of chemicals not only threatens the health of bees, but also threatens human health by leaving residues in bee products.

As with many living species, there are many diseases, parasites and pests that affect honeybees (*Apis mellifera* L.) and reduce their survival and productivity. All honeybee infectious and parasitic diseases are contagious, and if precautions are not taken, they can spread from colony to colony within the apiary, between apiaries, between regions due to migratory beekeeping, and between countries due to queen and colony trade, and may affect the whole world's beekeeping in a short time. (1,2).

Depending on the density of the varroa mite, which was determined to have entered our country from Bulgaria in 1976, our colonies began to be more affected by bacterial, fungal, viral diseases and parasites (3, 4). Various substances and drugs have been used unconsciously for years to combat parasites and diseases (5,6,7,8,3,9,10,11). As a result of unconscious use, varroa lines that are resistant to these drugs are formed, the effectiveness of the drugs used decreases and drug residues become an important problem in terms of food safety and human health.

Today, chemical, genetic and cultural control methods are intensively studied in the fight against honeybee diseases and parasites. Especially in the fight against parasites, chemical control method is the most preferred application by beekeepers because it is easy to apply and its effect is seen immediately. However, it has been proven in many studies that disease and harmful factors have begun to gain resistance against the chemicals used in recent years (7,8,12,13). In order to overcome these problems, studies on the development of bee lines resistant to parasites and infectious diseases have been accelerated in many countries such as America, European countries and Australia.

At this point, hygienic behavior is the primary mechanism of resistance to disease and parasites. For this purpose, efforts are being made to develop lines with high hygienic behavior characteristics in different honeybee genotypes (14,15,16,17). While efforts to develop resistant lines continue, the use of substances that are not harmful to human health for the control of diseases and pests has become a subject that has been intensively studied.

While Varroa lived symbiotically in the *Apis cerana* (Indian Honey Bee) bee until the 1960s, it began to live parasitically in the *Apis mellifera mellifera* after the 1960s. The transition to a parasitic life has caused deep harm to honey bees and even caused many colonies to disappear in some years.

1. Biology of Varroa

Varroa is a very dangerous external parasite that lives on larvae, pupae and adults of honey bees, feeds by sucking their blood fluid (haemolymph) and multiplies unnoticed in the colony for a long time (1,18,19). Adult female varroa vary in color from dark red-brown to reddish brown and have a transverse oval shape (Figure 1). It is on average 1.0–1.8 mm long and 1.5–1.2 mm wide.



Figure 1: Female Varroa.

Males are smaller than female varroa. Mouth parts are piercing and sucking. There is a pair of sharp, curved needle-like cliceres (chelicera = piercing organ) in the mouth. The mite clings between the segments (body parts) of the bee with the help of hook-shaped needles located in the front part of the cysts. There is a pair of long, mobile pedipalps on each side of the cliques. These enter the wound opened by puncturing the cuticle. Varroa has 4 pairs of legs and its legs are 6-segmented, short, strong and thick. There are suction cup-shaped lobes at the ends of the legs that enable adhesion (20).

They breathe through trachea thanks to their developed tracheal system. The ends of the tracheal tubes, which are distributed in the form of a network within the body of the mite, open to the outside with a hole called “stigma”. The central nervous system consists of nodal nerve fibers located above and below the esophagus (20). The reproduction of the mite begins in the spring with the development of bee larvae and continues until the last young worker bees emerge in the autumn (figure 2,3,4).



Figure 2: Varroas in the workerbee cell.



Figure 3: Varroas on the workerbee



Figure 4: Varroa on the workerbee abdomen.

In other words, when the queen bee completes its egg-laying function, the parasite also stops laying eggs. As the food given to the bee larvae in the honeycomb cells increases, the temperature rises and the male eyes with offspring are seen, the reproduction of the parasite accelerates. Once mated, the female parasite stores the sperm of the male parasite in the spermatheca sac and then uses these spermatozoa to fertilize the eggs. Mating takes place inside the honeycomb cells before the adult bees open the eye and emerge.

Since the mouth structures of male parasites are developed only for mating, they cannot feed and die shortly after the honeycomb eyes open. Fertile female parasites enter the eyes, where 5-6 day old larvae are developing in the spring, 15-45 hours before these eyes are closed. More than one mite can enter the same eye (21,22,18,19). Female parasites that feed on the blood of adult bees are not capable of laying eggs, so female parasites must feed on larval blood for 4-5 days in order to lay eggs. It enables the development of the ovaries of the juvenile non-hormonal parasite found in the larval blood.

The female mite, whose ovaries have developed, lays her first egg 60 hours after the eyes are sealed and continues to lay eggs at 30-hour intervals from then on. It is reported that the first egg is unfertilized ($n=7$ chromosomes) and the next eggs are fertilized ($2n=14$ chromosomes) (19). According to research, the female mite completes its full development in approximately 8-10 days, while the male mite completes its development in 6-8 days and adult individuals emerge. Six-legged larvae hatch 24 hours after the egg is laid in the closed eyes by the female mite. Generally, 2-3 female parasites can become adults in worker bee cells and 3-5 female parasites can become adults in queen bee cells. In queen bee cells, the queen completes its development period and leaves the eye before the mite becomes an adult.

In the honeycomb cell, the old female and the newly mated young female mite, they stay in the honeycomb cell until the young adult bee leaves the eye and they leave the eye together with the bee. Young female mites try to find a suitable comb cell again after 4-13 days to lay eggs. The lifespan of female varroa is 2-3 months in summer and 5-8 months in winter. Mites live on worker bees without laying eggs during the winter months when there is no brood chamber in the colony (21,22,20). It has been determined that bees lose a significant amount of body protein as a result of Varroa sucking blood fluids (6,23). This situation negatively affects the survival and longevity of the bee, especially in the winter season. When the bee it is feeding on dies, the mite leaves it and looks for

another host. It may also be possible to see young female mites walking on the honeycomb, looking for a new host in the hive and trying to choose a suitable honeycomb eye to lay eggs.

They usually attach to the bottom of the bee's abdomen and settle between the segments. Varroa transmission from bee to bee also occurs on flowers while bees collect pollen and honeydew (nectar). The parasite does not prefer other Hymenoptera (membrane winged animals); For example, bumblebee (*Vespa crabro* L., *Vespa orientalis* L. = bee monster) and yellow bee (*Polistes* spp.) are reported (24,25). Varroa is first seen more frequently in colonies with old queen bees that lay drone eggs, then in weak colonies, in strong colonies that are marauding, and in colonies that are preparing to give birth. When the contamination rate reaches 20-30%, the pest becomes visible. However, it is very difficult to remove mites from hives of this density. In fact, detecting the bee mite early and applying the necessary treatment methods without delay is very important in terms of preventing the spread of the pest and saving the colony with minimal damage.

2. Detection of Varroa Infection in Honey Bee Colonies

Around 200-250 adult bees are taken from each colony infected with Varroa and left in jars filled with powdered sugar. The jar is shaken to ensure that the bees get into the powdered sugar (26).

The jar is emptied into a strainer and the bees and spilled varroa are counted.

The percentage of varroa contamination is calculated according to the formula below.

$\% \text{ Varroa. Infection Rate} = \text{Total Varroa number} / \text{Total Worker Bees} \times 100$ (27).



Figure 5: Contamination detection with powdered sugar.



Figure 6: Pouring powdered sugar into a strainer for varroa counting.

3. Control of Varroa

3.1. Chemical Control

To date, chemicals have been used mostly in varroa control and have been successful up to a point. Due to the reproductive biology of bees and the parasite, chemicals did not eliminate the parasite, but their continued use helped keep it below the economic harm threshold. The chemicals that have been used extensively until today should be used with caution, both because the parasite can become immune due to their constant use and because they accumulate in bee products and threaten human health (5,28,7,8,3,12,10,27).

3.2. Biological Control

Considering that Varroa prefers drone eyes, the number of mites can be significantly reduced with biological methods. For this purpose, one or two half-empty frames with a 5-6 cm honeycomb piece attached to the top are placed in the middle part of the colony. Worker bees immediately start knitting male eyes into these half frames. The larvae hatching from the eggs laid in the eyes are 5–6 days old, and when the eyes are completely closed, the frames provided are removed and the combs are destroyed (27,26). If this application is done 3-4 times and at the end of the honey collection period when the colony least needs young worker bees.

The number of varroa in the hive can be significantly reduced. Likewise, biological control is carried out by placing the combs containing the male offspring's eyes into the hive and killing the mites by removing them after the eyes are closed (29,2,30,31).

3.3. Physical Control

Physical struggle; It is done by changing the environmental conditions in which the pest lives within limits that they cannot tolerate for a certain period of time. When the hive temperature was artificially increased to 46-48 °C in a controlled manner, it was determined that the mite on the adult bee body could not survive any longer, and this method began to be used in some special hives. However, its use in practice has not yet been possible (30).

3.4. Legal Combat

Special laws and regulations have been prepared to prevent plant and animal diseases and pests that can enter from outside through various means or to prevent them from spreading from contaminated areas to clean areas within the country. The process of managing and suppressing all kinds of internal quarantine measures and investigations, with certain laws and regulations, about diseases and pests that are not desired to spread and contaminate is generally called "Legal Struggle". For this purpose, the "Beekeeping Regulation", which was last published in the Official Gazette No. 25118 on May 25, 2003, stipulates that domestic bee transfers are prohibited in case of the presence of diseases and parasites in the bees such as varroa, bee lice, nosema, American Foul Rot, European Foul Rot and Lime Disease. It states that struggle is mandatory and that the necessary permission can be given for transfer to hives that have been combated and treated for this purpose (32).

3.5. Genetic Control

In recent years, studies on breeding bee lines resistant to parasites and infectious diseases have been accelerated in many countries such as America, Europe and Australia.

Bee Genetics experts state that different bee races show different sensitivity to the varroa parasite, and that there are colonies with different sensitivity even

within the same race. The selection made on this subject is based on the queen bees raised from colonies resistant to the parasite.

They obtained findings that colonies were more resistant to the parasite. It is stated that if these studies are continued and put into practice, there will be a great improvement in the fight against varroa (14,15,16,17).

4. Use of Organic Acids in Varroa Control

In order to overcome residue problems in bee products, natural substances that do not have harmful effects on food safety and human health have recently begun to be sought for varroa control. As a result of these studies, formic acid, lactic acid and oxalic acid have become the preferred natural substances in terms of suitability for the purpose (32,33,34,27,26).

4.1. Use of Formic Acid

It is a colorless, volatile, and weak organic acid. Its evaporation feature is used and it gives successful results when the air temperature is between 10-25 °C. During the application period, hive entrances and ventilation holes must be fully opened. Many effective application methods have been developed for use in bee colonies. It is easy and practical to use, and the dose adjustment must be made very carefully according to the colony size and hive type. It is generally recommended to use it in early spring or late autumn when there are relatively fewer broods in the hive and the temperature is suitable. Since evaporation will be less at low temperatures, the effectiveness will be less, and at high temperatures, evaporation will be more, which may cause bee deaths and queen bee losses in the colony. Although formic acid is naturally present in honey, the application must be completed 6–8 weeks before the honey harvest to avoid quality problems (32,35,36,31).

As a result of treatment with formic acid for more than 14 days, it can kill up to 95% of parasites. It has also been observed that some varroa species in cells with closed offspring die in cases where there are closed offspring in the frames. Formic acid is naturally found in small amounts in honey. Treating hives with formic acid can increase these levels. The detection threshold of the residue in honey is at the level of 150–600 mg/kg, depending on the lightness of the flavor of the honey. If formic acid is applied in autumn, the residues in the stock

honey in the hive will be 40-200 mg/kg in the next spring, while in fresh honey it will be 25-50 mg/kg, depending on the type of honey. If formic acid is used in spring, the residue amount may be higher (32).

4.2. Lactic Acid Application

Minus is a sharply sweet and weak organic acid, produced from lactose by some bacteria. Lactic acid is used to give some foods a sour taste and is also used to preserve foods. Lactic acid is a substance widely found in nature. It occurs when energy is burned in the human body. It is also found naturally in foods such as honey, sour milk, cheese and yoghurt. Since lactic acid application is a labor-intensive application, it is generally preferred by beekeepers with a small number of colonies (32,35,31).

The best results in lactic acid application are obtained when lactic acid is sprayed on the bees and the walls of the hive (32,35). Although it is recommended to use it in early spring and late autumn, when the environmental temperature is between 7-30 °C during application, when there is no honey in the hive, there is no harm in using it in all periods except the main nectar flow period and honey harvest period. It should be preferred for application, especially in the afternoons and on windless days when field bees return to the hive. Using sprayers that produce very fine sprays for spraying gives successful results. The most appropriate dosage is to spray 5 ml of 15% acid solution on each side of a frame completely covered with bees.

It is reported that 3-4 applications twice a year, 3-4 days apart, will be sufficient, depending on the shedding status of Varroa (32). It takes a lot of time as it has to be applied to all frames in the hive one by one, and therefore it is recommended for beeyards with a low number of colonies, natural and artificial offspring and brood hives. Its effectiveness can be 80-95% in broodless colonies. Since it has no effect on the varroa species in the closed honeycomb cells, its effectiveness decreases significantly since the majority of the parasite (70-80%) will be in the closed eyes during periods when the brood is dense in the hive. It is recommended to monitor the effectiveness after spraying and to repeat the application or try another application if the parasite density does not fall below the economic threshold.

After 3 applications made three days apart, the lactic acid accumulated in the honey stored in the honeycomb increases from 200 mg/kg to 1500 mg/kg. How-

ever, most of this amount decreases to normal levels within 7-8 weeks following the application. Lactic acid in flower honeys.

It is naturally found between 40 and 400 mg/kg, depending on its source. A person who consumes 30 g of honey per day receives approximately 6 mg of lactic acid, while a person who consumes 180 g of yoghurt receives 1800 mg of lactic acid. If the use of lactic acid is not allowed during the main nectar flow and honey harvest periods, lactic acid does not pose any residue risk in honey (32). Honey, It contains both levulose (fruit sugar) and dextrose (tea sugar) forms of lactic acid. Commercial lactic acid is a mixture of these two forms, but pure or naturally occurring lactic acid is equally effective in combating varroa.

4.3. Oxalic Acid Application

Oxalic acid is a strong acid and can irritate tissues. Extremely high doses are lethal. Oxalic acid is a compound that occurs naturally in many plants. The human body also synthesizes oxalic acid from ascorbic acid. Oxalic acid can form salt forms with calcium, iron, sodium, magnesium or potassium, known as oxalates. Oxalates are in the form of sharp-edged crystals that are slightly soluble in water and thus irritate the tissue. For this reason, diets containing large amounts of oxalic acid or oxalates cause irritation in the digestive system, especially the stomach and kidneys. Oxalates also occur naturally in plants (32,39,35,31).

Oxalic acid is generally used once a year in late autumn, when there are few or no offspring in the colony, on a windless day and in the afternoon when the ambient temperature is between 7 and 30 °C (32,35). More than one application may increase bee mortality or slow down colony development next spring. Dosage adjustment is very important in oxalic acid application. For this, 1 liter of boiled and cooled clean water is mixed with 1 kg of sugar and 75 g of crystalline oxalic acid di-hydrate is added to this mixture. In this way, a 3.2% solution is prepared. It is taken with the help of a large-scale syringe and applied to the bees between the frames by dropping 5 ml onto each frame completely covered with bees. The solution taken into the syringe can be dropped on the bees between the frames, or it can be sprayed on the bees with the help of a sprayer. If excessive amounts or high concentrations of oxalic acid are used, it can kill bees or their brood.

For this reason, attention should be paid to the amounts used when preparing sugar syrup and oxalic acid solution. Oxalic acid can be easily used in all

types of apiaries as it does not require much labor and is reliable in terms of food safety and human health. Since it only kills the parasite on adult bees, it should be used in periods when there are few or no broods in the hive. While it can show 95% effectiveness in periods when there are no offspring in the hive, this effect is around 40% in periods when there are many offspring. After the use of oxalic acid, the status of the parasite in the colony and the effectiveness of the drug should be monitored. If the parasite has not fallen below the economic damage threshold, a second oxalic acid application should not be preferred and another drug should be applied instead (32).

Oxalic acid is naturally found in low amounts in honey. When it reaches high levels (400–900 mg/kg), it can create remarkable flavors in honey. When applied once in autumn or early spring, it does not leave a significant residue in honey (41).

5. Conclusion

Today, it is not possible to completely destroy varroa living on adult bees or inside their closed brood cells with current methods. In order to have efficient beekeeping and produce healthy bee products, it is necessary to reduce the density of varroa in the hive. For this purpose, it is necessary to apply continuous treatment at certain periods. The main goal is to reduce the number of varroa in the colony to a level where it cannot cause harm (40). For this purpose, physical, biological, genetic and chemical control methods must be used together in harmony. Currently, the least labor intensive, cheapest and easiest to apply method among these methods is chemical control. Many chemicals and natural substances have been used for this purpose from the time when it became necessary to fight against Varroa until today.

However, the majority of these chemicals were applied unconsciously and incorrectly without a license, resulting in decreased effectiveness and the emergence of resistant varroa lines. Accordingly, since bee products, especially honey and wax, are absorbent substances, residue problems have arisen in the products (16). Especially in the European Union countries, as these problems began to be perceived by the public, the pressure on management units to solve these problems began to increase. As a result, developed countries that consume bee products demand residue-free products. For this reason, we have to produce

products that do not cause residue problems, both for sales and for our own health. The European Union has kept under surveillance the drugs and various chemicals used in the fight against varroa in order to ensure unity among member countries while establishing organic agricultural production standards. As a result of scientific studies on these substances, formic acid, lactic acid and oxalic acid have been suggested as the only substances to produce organic and healthy products in beekeeping (32,39,36,31).

These organic substances are naturally occurring substances in honey, although their amounts vary depending on the nectar source. These substances do not have any harmful effects when used according to their directions. In addition, when the application time is taken into consideration, the amounts in honey remain within natural limits. For this reason, application should never be made before the honey harvest, and applications should be made in early spring and late autumn. On the other hand, there is no evidence that varroa has become resistant to these substances (32). Ensuring the use of drugs alternately (for example, formic acid in spring, oxalic acid in autumn and lactic acid in the other spring) is also very important in preventing varroa from gaining resistance to these chemicals (7,8,12,13).

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DIAGNOSIS OF BEE DISEASES IN ADULT HONEY BEES AND BROOD ACCORDING TO CLINICAL FINDINGS

Metin GÜRÇAY

Introduction

Beekeeping is an agricultural activity in which honey bee colonies are used to maximize the number of field worker bees during periods when nectar flow is most abundant, depending on the region where they are located, and to use these colonies for the production of honey, pollen, royal jelly and pollination of plants (1).

First of all, it should be known that beekeeping is a branch of animal husbandry that is almost entirely based on flora. Flora diversity and potential are essential for efficient cultivation. However, beekeeping is a sector that requires the use of all kinds of technology and scientific developments as well as being a skill job. In this sense, in today's beekeeping, improved, productive, disease and parasite resistant bee lines and strains are used, colony management regarding care and feeding is carried out in a planned and programmed manner, migra-

tory beekeeping opportunities are evaluated and the production of healthier and more qualified bee products is prioritized (1).

Honey bees are creatures of genus *Apis mellifera* (*Apis mellifera* means honey collector) in the arthropod phylum in the animal kingdom, in the membrane winged order. The body structure of honey bees consists of 3 parts: head, thorax and abdomen. Honey bees are social creatures that live in colonies. There are three different individuals in a bee colony. These individuals differ in morphological, physiological and behavioral aspects. Under normal conditions, colonies contain 1 queen bee, tens of thousands of worker bees, the number of which varies depending on the season, and 0-5000 male bees, the number of which varies depending on the season and need (1).

The honey bee life cycle begins as an egg in all individuals. The egg stage is followed by the larva and then the pupa stage. Finally, an adult bee emerges. The queen bee becomes an adult in 16 days, the worker bee in 21 days, and the drone in 24 days. Queen bees can live for 5-7 years

While the average lifespan of worker bees is 1-1.5 months. Queen bees and drones weigh an average of 200 mg, while worker bees weigh an average of 120 mg. While queen bees have a long and narrow body structure, the body of male bees is a little more plump. Bee colonies are communities that live with a certain hierarchy and order. Female individuals do the main work in the colony. Male bees have no duties other than mating. The most important duty of the queen

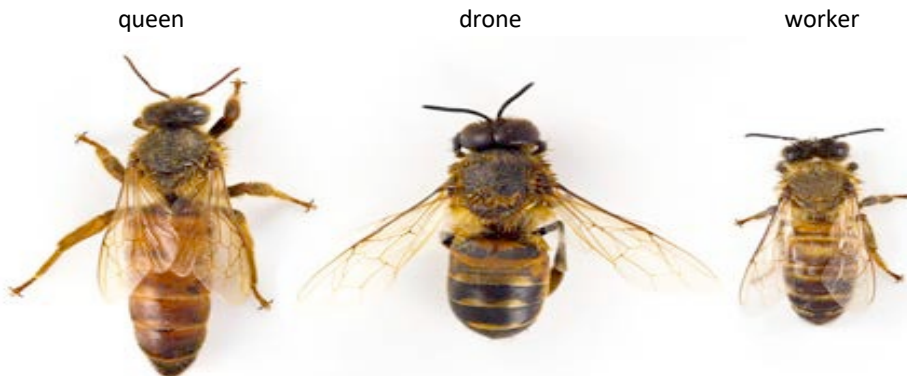


Figure 1: Clinically healthy *Apis mellifera* queen, drone and worker (left to right). Note the comparative difference in size and morphology (2).

bee, one of the colony individuals, is to ensure the continuity of the colony by laying eggs. A high quality and young queen bee can lay an average of 1500-2000 eggs per day. They also maintain order in the hive thanks to the pheromones they secrete. Baby feeding, royal jelly and wax secretion, honeycomb cleaning, All other work in the hive, including ventilation, defense, nectar collection-storage-ripening, pollen-propolis collection, water collection, and even the feeding and care of the queen bee, are carried out by worker bees (Figure 1), (1).

The embryonic development stages of the embryo, which is formed by the fertilization of the egg cell with spermatozoa, until the larva stage can be summarized in this way. Cell division is completed in 14-16 hours and blastoderm forms between 30-35 hours. Embryo membrane with mesoderm and mesenteron rudiments 42-48. It is completed within hours and in this period the respiratory system and some other differentiations begin to be seen. The development of the stomodeal and proctodeal sheaths and the neural canal is completed by the 66th hour, and all developments in the egg period are completed after the 76th hour. In the eggs attached to the base of the honeycomb by the queen bee within a three-day period; While there is no change in the outer part of the eggs other than their physical location, embryo development continues in the inner part. Once embryo development is completed, the shell of the egg cracks and larvae are formed. The time it takes for the eggs to become larvae, which will form all individuals in the colony, is equal and is approximately three days or 76 hours. After this stage, they continue to develop as larvae, prepupa and then pupa and form adult individuals (Figure 2), (3).

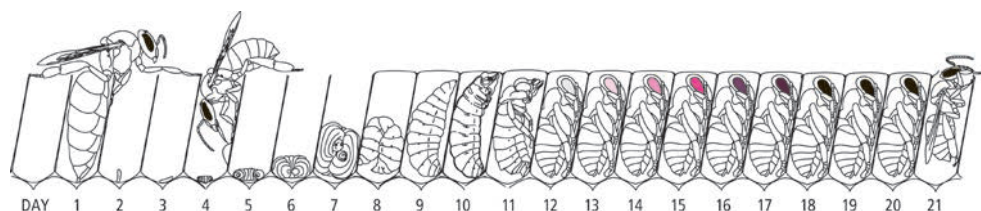


Figure 2: Clinical development of healthy *Apis mellifera* offspring. Honey bees undergo a complete metamorphosis from egg to larva and pupa until adulthood (2).



Figure 3: Clinically healthy puppy appearance (4).

The caps are uniformly brown, tan or cream. Except for a few cases where the cell lid is not yet fully constructed, each lid is slightly raised or convex, without any holes. If the developing offspring are healthy, the honeycomb cells are examined. The larvae are shiny and pearly white, with an orange gut line running along their backs. Healthy pupae under the caps are white at first but darken in color as they develop into adults. The eyes start to get colored first. Physiological developments seen in honey bee colonies and the encounter of honey bees with epidemic diseases at a stage of their life cycle negatively affect the development and progress of beekeeping activities. Due to these epidemics, the clinically continuing physiological life affects adult bees and breeders, causing sudden death and shortening of the normal life span of the bees, causing colony losses and decreasing the amount of production per unit hive. Clinical examination of bee diseases is an important part of beekeeping. Beehive inspectors and beekeepers should be able to recognize bee diseases. In this book chapter, the diagnosis of disease agents in bees will be studied based on the clinical symptoms specific to the disease agent in adult bees and developing bees. When these diseases are recognized clinically, early recognition of the damage to the colony or hive will reduce the damage caused by these diseases (Figure 3), (1).

Diseases seen in honey bees and their diagnosis according to clinical symptoms

American foulbrood disease (AFB)

This disease is a very dangerous and epidemic brood disease that occurs in honey bee larvae and causes the larvae to die and stink. The larvae of all three individuals are affected by the disease. The name of the pathogen has been defined as *Paenibacillus larvae* in recent years. *Paenibacillus larvae* is a gram-positive bacterium. Bacterial spores transmitted to the larvae along with the food cause disease in the larvae. These spores can survive for 33 years in the hive, 60 years in the soil and 45 years in the foundation comb. Although the distribution of offspring in the brood areas in healthy colonies is frequent and uniform, it is seen that open and closed eyes are mixed in the brood distribution of sick colonies. The emptied buds are abandoned after the dead larvae are discarded. The color of the diseased larva changes from white to yellow and then to brown. Eyes with closed eyes are concave instead of being convex and have a sunken, pit-like appearance. Additionally, some of the lidded eyes have pinhead-sized holes above them. If a matchstick is inserted into the eyelets and the baby remains are pulled out, it will be seen that the residue extends 4-10 cm. When the hive cover is opened, the smell of glue is felt. If the chicks have died during the pupation period, it is seen that their tongue hardens and stands upwards, dividing the honeycomb eye into two. The most accurate and effective method of combating this disease, which is notifiable in our country, is to destroy the diseased colonies by burning them completely. The hive body should be burned by passing it through the flame thoroughly with a blowtorch, and contaminated materials such as bellows, masks, hand irons, feeders, and queen bee grills should be disinfected. This prevents the disease from being transmitted to other colonies. Antibiotics containing sodium sulfathiazole and oxytetracycline hydrochloride (OTC) are used in the treatment of the disease. Antibiotics are effective only on vegetative forms. Spores cannot be killed with these. Therefore, antibiotic administration does not help. The best method to prevent the disease; When purchasing colonies, check whether there is any disease and sick colonies should not be purchased. Beeswax used in honeycomb making must be sterilized at 120 ° C for 10-15 minutes. For this reason, care should be taken to purchase the basic honeycomb from companies that have a production permit (Figure 4, Figure 5), (5).

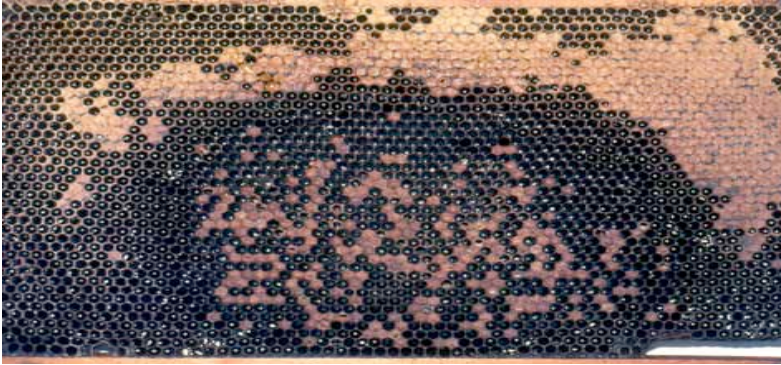


Figure 4: Perforated caps characteristic of American foulbrood on honeycomb. Some closed eyes appear to have been pierced with a needle, their colors are faded, and they appear sunken. When the hive is opened, the smell of heated glue and the smell of spoiled fish are felt. Closed puppy eyes are irregular (5).



Figure 5: Dead larvae change color from dull white to light and then dark brown. Since the dead larvae in the eyes are sticky, they grow 2-8 cm slimy when removed with a garbage can (5).

European foul rot diseases

It is another foul rot disease that is seen all over the world and in our country, except New Zealand. It was given this name because the first studies on the disease were conducted in Europe. The cause of the disease is a non-spore-forming gram (+) bacterium called *Melissococcus pluton*. Bee larvae take the pathogenic bacteria into their digestive system through the food carried by feeder bees. Bacteria that settle in the digestive system of the larva develop in the intestine, and the disease agent is excreted into the honeycomb chamber with feces after the fry enter the pupal period. While worker bees clean these residues from the honeycomb cells, they transmit the disease to healthy larvae. Adult bees that are carriers are not

affected by this disease. The disease is usually seen at the beginning of the nectar season, when brood rearing is intense, between May and June. The disease occurs in nectar shortage and cold weather conditions. When the amount of nectar coming into the hive reaches the highest level, the severity of the disease decreases. In hives infected with the disease; In the incubation area, the area with the young is irregular and messy. The larvae become infected with the disease when they are 3-4 days old. Deaths usually (90%) occur in squints. The diseased larva acquires a yellowish color, then the color changes to brown. The dead larva sometimes remains upright in the honeycomb chamber, and sometimes remains attached to the bottom of the chamber in a melted form. If death occurred after the eye was closed, the eye collapsed, was pierced, and became discolored. Larvae that are dead but cannot be expelled by worker bees rot. There is little or no stickiness and elongation in the decomposing larva. Rotten larvae have the smell of putrid meat. The first precaution against the disease is to keep the hives strong. Because the disease causes the greatest damage in weak hives. Since there is a direct relationship between the emergence of European foulbrood and colony stress, measures such as avoiding practices that may be a source of stress for colonies, feeding with sugar syrup when necessary, not creating pollen deficiency in the colony, and working with young queen bees prevent the emergence of the disease or reduce the negative effects of the disease. . Oxytetracycline and group antibiotics and erythromycin can be used in the treatment of the disease. Drug treatment should be completed 3 weeks before nectar flow. It should not be forgotten that unconsciously and randomly used antibiotics will cause increased resistance in bacteria and residue in honey (Figure 6, Figure 7), (6).



Figure 6: Appearance of European foulbrood in the comb (6).



Figure 7: Appearance of European foulbrood in the comb (7).

One of the most important symptoms is that the larva turns from its normal dull white to a dull yellow, then brown, and then black. The disease has a characteristic sour, brewer's yeast, or rotten fish-like smell. Generally, larvae die within 4 days after hatching. Deaths occur before the comb cells are closed. While American foulbrood is more common in closed broods, European foulbrood is more common in open broods. Since dead larvae are less sticky, they can be cleaned from the eyes by bees when they dry. In colonies infected with the disease, anxiety, reluctance, laziness and sensitivity are noticeable. Larvae usually die when they are in crescent form (6).

Chalkbrood diseases

Lime disease is a juvenile disease caused by the fungus called *Ascosphaera apis*. It was first diagnosed in Turkey in 1988. There are three subspecies of the fungus. *Ascosphaera apis alvei*, *Ascosphaera apis minor* and *Ascosphaera apis major*. *Ascosphaera apis* is a fungus with a heterothallic structure [mycelia having different sexes as male (+) or female (-)]. Hyphae from two different sexes combine to form spore sacs approximately 47-140 μ in length. The inside of these sacs is filled with spores of the fungus. The larva ingests *Ascosphaera apis* with food. Larvae are more susceptible to diseases when they are 4-5 days old and a few hours after the honeycomb eyes close. Spores are very durable. They can maintain their disease-causing ability for 15 years.

It is most common in spring and autumn. Strong colonies can overcome the disease during the summer months. Spores that remain alive on the honeycombs for years without causing disease become active again when they find suitable conditions for reproduction. Male bee larvae are mostly in the side combs of the hive and generally stay below the incubation temperature. For this reason, the symptoms of the disease are first seen in the drone larvae on the edges of the combs and in the edge combs. Because the most suitable temperature for the development of fungi is around 30 °C. In the later stages of the disease, white mummified larvae are seen in the middle parts of the brood area and in the middle of the honeycombs. If the larvae are infected with only (+) or (-) sexual mycelia, they are white as chalk. If both (+) and (-) sexual mycelia are infected, the color of the diseased larva becomes grayish black. Environmental pollution, intensive use of antibiotics, feeding bees with artificial nutrients, excessive humidity, and the use of adulterated wax are effective in the formation of the disease. During the colony checks carried out by beekeepers in early spring, especially on days when the weather is overcast and cold, the temperature of the brood in the hive decreases, the resistance of the brood decreases, and the development of the fungus is activated as more oxygen penetrates into the larval tissues with the chill (Figure 8, Figure 9), (8).



Figure 8: Appearance of Chalkbrood disease on the honeycomb (9).

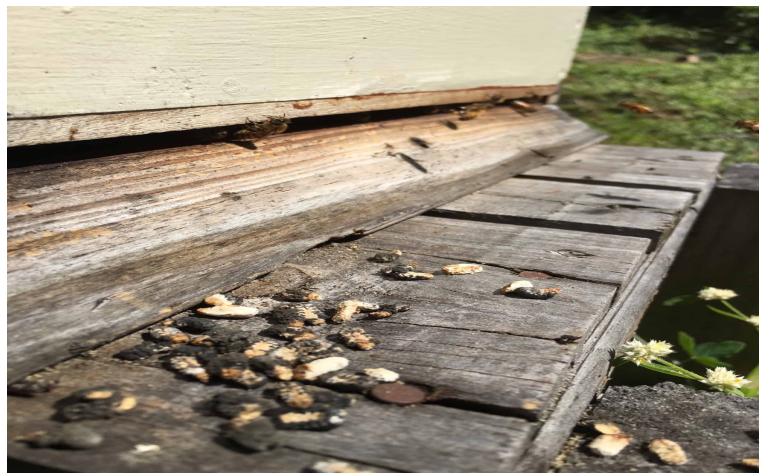


Figure 9: Appearance of Chalkbrood disease at the hive mouth (10).

Sacbrood diseases

It is inherited. The larva is called ‘crazy larva’. Looting, honeycomb exchange between hives, and food exchange are important to prevent the spread of the disease. Death occurs in bees before they enter the pupa period. Larvae are most susceptible to disease when they are 2 days old. During this period, it is important that the larval food carries the virus. The infection disappears towards the winter season. The virus continues to reproduce in adult bees that do not show symptoms of the disease but have the virus in their bodies. Young worker bees are the most prone group to encounter the virus and get the infection because they clean the comb cells. Young workers acquire the infection when they remove dead larvae with sacbrood, even if they ingest a very small amount of the fluid in the sac. The virus reproduces in the hypopharyngeal glands of the bees and reaches large numbers within a day. It is transmitted to other honey bees through food transfer and gland secretions. The shortening of life expectancy in the colony causes the severity of the disease to increase. Sick bees stop eating pollen. They become field bees earlier than normal. They collect much less pollen than a healthy adult bee should. Due to the infection, changes occur in many tasks in the colony. Sacbrood is most common in spring and early summer. During the large nectar flow period, the number of field bees increases and the disease heals on its own.

The virus then overwinters only in infected adult bees. Many of these bees live shorter lives, and the number of infections decreases in the new season. The larvae are normal until the honeycomb eyes are closed. Sick larvae appear as bags filled with water. It can be easily taken out of the honeycomb chamber by holding it with a needle or forceps. The white color of the sacbrooded larvae inside the comb cell later turns into a pale yellow color and cannot enter the pupa period. The head of the infected larva faces the comb eye and dies in an upright position. In the disease, the head of the larva turns black first. The larva is in a bag filled with a greenish liquid. This is why it is called bag rot. Then the larva dries out, its color turns dark brown and it dies after a short time (figure). The combs have a puzzle-like appearance, as in European foulbrood. Generally, dead bees do not have a distinctive smell. There are 2 holes in the diseased comb eyelids (Figure). Liner pieces belonging to dead larvae can be easily removed by young worker bees. A strong bee colony can easily clean all cells containing dead larvae. There is no chemical cure for bag rot. The damage of the disease can be reduced with healthy protective measures. It is necessary to keep the hives in damp places, preventing them from getting moisture, especially from the bottom. For this purpose, hives should be placed at a height of 30-40 cm. Replacing the queen bee and strengthening the bee family will help treat bag rot (Figure 10, Figure 11), (11).



Figure 10: Sacbrood. Perforated Covers (12).



Figure 11: Progressive view of larval decay (13).

Black queen cell virus diseases

Black queen cell virus belongs to the Dicistroviridae family. Black queen cell virus, Filamentous virus and Y viruses were first detected together with *Nosema apis*. Black queen cell virus was first detected in sealed queen rings with dead queen larvae or pupae. Dead larvae and pupae appear in colors ranging from dark brown to black. Sick larvae appear light yellow in color. It has hard bag-like skin. In this form, it resembles patients with sacbrood. Black queen cell virus does not multiply when young worker bee larvae are eaten by young worker bees. The Sacbrood-like virus reproduces in large quantities when injected into the pupa (Figure 12), (14).



Figure 12: Clinical view of the effect of black queen bee virus on the larva (15).

Adult Bee Diseases

Acute bee paralysis virus diseases

Acute bee paralysis virus belongs to the Dicistroviridae family. Acute bee paralysis is caused by the acute paralysis virus. Acute paralysis virus is found in seemingly healthy bees, especially in the summer. If Varroa destructor is present together with acute paralysis virus in the colony, deaths occur in adult bees and larvae. When Varroa feeds on the hemolymph of bees that appear healthy but carry the virus, it causes the virus to multiply. Thus, the virus rapidly increases to a level that has a systemic effect and shows its lethal effect. It acquires varroa virus by feeding on bees infected with the virus. It acts as a vector for Varroa virus and transmits it to other adult bees, larvae and pupae. Acute paralysis virus reproduces rapidly in the body of young larvae and adult bees and causes severe disease in the colony. The number of adult bees decreases rapidly (Figure 13),(16).



Figure 13: Clinical appearance of acute bee paralysis (17).

Chronic bee paralysis virus diseases

Chronic bee paralysis is caused by the chronic paralysis virus. It was first described in the United States in 1809 in an article written by Huber. Burnside demonstrated the disease in 1933 by giving viral material from infected hives to healthy bees. The virus measures 30-65 μm . Chronic paralysis virus is in satellite form. The bee's immune mechanism prevents the chronic paralysis

virus from multiplying. Chronic paralysis virus can be confused with acute paralysis virus. Bee paralysis is more common in temperate regions. Sick adult bees are noticed with their wings open, drooping and trembling. Bees cannot fly, they are found in clusters crawling in front of the hive. Sick bees in the hive gather at the top of the frames with brood. When smoke is introduced into the hive, infected bees do not move. Their abdomens are swollen because they cannot empty their intestines. The reason their bellies are swollen is because their honey sacs are full. This condition can cause dysentery. Paralyzed bees lose their body hair and their abdomens appear dark and shiny. Because there are no feathers, the abdomen appears smaller than normal. Paralyzed bees coming from flight are attacked by healthy bees and are not allowed into the hive. Sick bees that stay outside for 1-2 days begin to tremble, and 150-200 infected bees gather in front of the hive and die after a while. Abnormal wing tremors and sometimes thousands of bees gathering are important in the disease. Patients die within a few days of the onset of symptoms. Severely infected colonies may disappear suddenly. This situation occurs especially in the middle of summer. Hot and dry weather increases the severity of the disease. One of the typical symptoms of the disease is the queen bee leaving the hive along with a large number of worker bees (Figure 14), (16).



Figure 14: Clinical appearance of chronic bee paralysis in the colony (16).

Deformed wing virus diseases

Deformed wing virus was first isolated from adult honey bees in Varroa-infected bee colonies in Japan. In colonies with the disease, the wings of the newly emerging bees from the honeycombs are seen to be poorly developed or deformed. The virus is expelled with all the extracts of the diseased bees. It has a distant serological link with Egyptian bee virus. After the bees are infected with the virus, the wings of the bees inside the honeycomb chamber cannot fully develop and their shape is distorted. The young die during the larva and pupa stages. If adult bees are infected with the virus, they appear normal until they die. The virus is lethal even when there is no varroa in the colony. Varroa species transmit the deformed wing virus to healthy bee offspring (Figure 15), (16).



Figure 15: Appearance of the wing disorder caused by Deformed Wing Virus (18).

Nosema disease

In 1909, German scientist Enoch Zander named the parasite *Nosema apis*, which infects the midgut epithelial cells of honey bees. In 1919, White named the infection caused by *N. apis* in honey bees as nosema disease. The disease occurs in both *Apis cerana* and *Apis mellifera* races. The disease was first transmitted to the Eastern honey bee *Apis cerana*, and then to *Apis mellifera* (Figure 16), (19).



Figure 16: Paralyzed intestine and healthy intestine in the nose (20).

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THE USE OF BEE PRODUCTS IN ANIMAL PRODUCTION AND NUTRITION

Sirac YAVUZ

Introduction

As the world's population is growing rapidly, the demand for protein-rich foods is increasing continuously. As a result of this increasing demand, the resources required for human nutrition on the one side and the resources required for the production of animal feed on the other side are under severe pressure. Consequently, there is an intense competition between these two requirements.

The direct consequences of this market competition are quite concerning. Primarily, environmental degradation is becoming a major problem. Environmental impacts such as the expansion of agricultural land, deforestation, overuse of water resources and the decline in biodiversity pose serious threats to a sustainable future. Furthermore, the rapid depletion of natural resources means that we will struggle to meet the needs of both present and future generations.

In order to cope with these challenges, we need to rethink the way we feed animals and find innovative solutions. Sustainable practices need to be adopted that can balance the needs of both humans and animals. For instance, the use of alternative protein sources such as insects in animal feed, optimising feed production through biotechnological methods and recycling waste can play an important role in achieving this balance.

Developing societies evolve their eating habits towards natural products. Financial situation is one of the most important factors in the consumption preferences of the society. The accessibility to all kinds of flavours and new products, foods containing intense and multiple contradictory nutrients contrary to human nature direct societies, especially those with a culture of nutrition, to the flavours they are essentially accustomed towards. Organic or natural honey is one of the first five products that come to mind when we think of bee products. The fact that it is very important in human nutrition and has survived intact throughout history is an indication that this miraculous food is a source of healing. The consumers interested in honey have contributed to the popularisation and recognition of other bee products. Bees (*Apis mellifera*) are organic product advocate pollinator insects that produce products such as honey, royal jelly, bee bread (perga), bee venom, bee pollen and propolis. In their own colony, these pollinators have different roles and responsibilities. For instance, drones are responsible for mating with the queen, while other female worker bees carry out maintenance and feeding responsibilities (Figure 1).

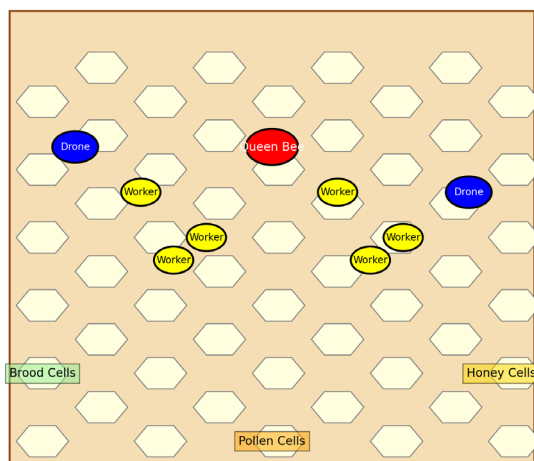


Figure 1: A hive of honey bee colony

Bee colony and products

- **Queen Bee:** Usually found near the centre of the hive.
- **Worker Bees:** They take on different tasks in various parts of the hive.
- **Drones:** Usually found in the upper parts of the hive, they are responsible for mating with the queen, they are physically distinct.
- **Honey combs:** The place where honey is stored.
- **Larval Cells:** Cells where the young bees are reared.
- **Pollen Stores:** Where pollen is stored.
- **Queen Cell:** The cell where the queen bee is reared, the part where royal jelly is fed.
- **Propolis:** Propolis is used throughout the hive, for sealing cracks.

Worldwide, there are approximately 107 million beehives. Among them, this number is distributed in various regions and the number of hives in each region varies. Below are the regions and the estimated number of hives:

Africa: 28 million

Asia: 46 million

Europe: 17 million

Latin America and the Caribbean: 12 million

North America: 3.1 million

Oceania: It was reported to have 1 million beehives (2).

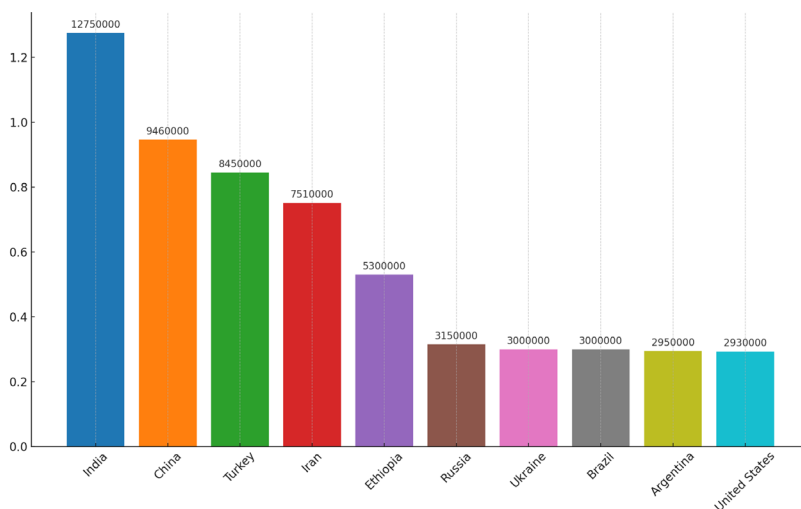


Figure 2: Worldwide beehives number (1)

Worldwide, beekeeping is widespread in 10 countries and Türkiye ranks in the top three (Figure 2). Due to its rich flora diversity and different climatic zones, Türkiye has very favourable conditions for honey beekeeping. Relative to other countries, Türkiye's climate diversity and agricultural areas provide bees access to floral resources throughout the year. Hence, this favours high productivity and quality in honey production. Nevertheless, management of water resources and environmental factors are important for the sustainability of honey beekeeping. Similar to other countries, beekeeping activities in Türkiye need to be protected against environmental sustainability and climate change impacts.

Bee Products

Products are natural substances produced by bees in their natural habitats and used by humans for various purposes. However, these products are substances that bees create in their life processes and are usually stored in their hives. Apiary products are used in different fields such as food source, health, beauty and industry. The following are the main bee products:

Honey

Honey is a natural flavouring with numerous pharmacological features. It contains numerous amino acids. Conventionally, it has been used in the treatment of various disorders including gastritis, diarrhoea, eye disorders, skin infections and wounds. Pure honey has been found to have antimicrobial, antioxidant and wound healing properties (3).

Propolis

Propolis has significant pharmacological potential, including antineoplastic potential (4). It was used in traditional medicine and found to exhibit antibacterial, antiseptic, anti-inflammatory and anaesthetic activities (5). It was investigated for its potential in the treatment of bacterial infections of the oral cavity (6). Furthermore, it has been found to have antiproliferative effects and has been recommended as a natural product for cancer treatment (4;7).

Bee pollen

A natural product obtained from the pollen of flowering plants, mixed with bees' digestive secretion enzymes and nectar and shaped by honey bees. Nowadays, it aims to become one of the widely used natural supplements. Pollen represents mainly the cells of the male reproductive spore of flowers and is the main food source of bees; it has concentrations of nutrients and phytochemicals and is rich in secondary metabolites. The ancient Egyptians described pollen as 'life-giving dust' in their time. Pollen is referred to as the only food with a wide range and the main biological components of bee pollen are polyphenolic compounds, phenolic acid derivatives and mostly hydroxycinnamic acids and flavonoid glycosides (8).

Royal Jelly

Nutrient-rich substance secreted by worker female bees to feed the queen bee. It has been studied for its potential health benefits, including anti-inflammatory and antioxidant effects. Traditionally, royal jelly has been used for various health problems, including skin and hair care (9). It is the most economically valuable product among all other bee products.

Bee bread (Perga)

Commonly known as perga, bee bread is a type of bee product that is rich in nutrients and is being researched for its potential health benefits. It is a combination of pollen and honey that bees produce and store in their hives. However, perga is known to contain a variety of nutrients, including proteins, carbohydrates and vitamins, which are essential for maintaining good health. Perga has traditionally been used in apitherapy, a form of alternative medicine that uses bee products for therapeutic purposes. It is widely believed to have anti-inflammatory and antimicrobial properties that may help in the treatment of various health conditions. Studies on perga are limited compared to other bee products such as honey, propolis and royal jelly. Nevertheless, studies have shown that perga may be beneficial for the immune system and may have potential applications in the prevention and treatment of some diseases (3;10).

Animal Production

Animal production refers to the processes of raising, breeding and managing animals for various purposes such as food, dietary fibre and other outputs. It encompasses a wide range of activities, including poultry production, livestock breeding, dairy production and aquaculture (11;12).

Animal husbandry involves the raising and breeding of animals such as cattle, sheep, goats and swine for meat, milk and other products. Livestock production is vital for food security and economic development, especially in rural areas where many people depend on animal husbandry for their livelihoods (13;14).

Poultry production focuses on the rearing and breeding of birds such as chickens, Türkiye and ducks for meat and eggs. In many parts of the world, poultry is an important source of protein and is often used in the production of processed foods (12;15).

Dairy production involves the rearing and breeding of dairy animals such as cows, goats and buffaloes for milk and other dairy products. In many countries, dairy farming is an important sector and provides a source of income and employment for many people (13;16).

Marine aquaculture, also known as fish farming, involves the rearing and breeding of fish and other aquatic animals for food.

Fisheries aquaculture is a growing sector globally, providing a sustainable source of protein and helping to reduce pressure on wild fish populations (12).

Animal production is a comprehensive and multifaceted industry that requires careful management and planning to ensure the health and welfare of animals as well as the quality and safety of the products they produce. All of the factors such as nutrition, health and environmental conditions play a critical role in determining the success of animal production operations (16;17).

Animal Production

According to FAO (18) data for 2020; Global milk production was reported as 887 million/ton, total meat production as 336 million/ton, chicken meat as 133 million/ton, pork as 109 million/ton and beef as 71 million/ton (Figure 3).

According to TURKSTAT (19) 2020 data; Türkiye milk production was reported as 23,158 thousand/ton, total meat production as 2,157 thousand/ton, chicken meat as 2,150 thousand/ton and sheep meat as 107 thousand/ton (Figure 3).

Milk production in Türkiye accounts for about 2.6 per cent of total milk production worldwide. This shows that Türkiye has an important place in milk production, but a relatively small share worldwide. On the other hand, Türkiye's total meat production accounts for only 0.64 per cent of the world total, suggesting a smaller role in meat production. In terms of poultry meat production, Türkiye accounts for about 1.62 per cent of world production. This indicates that Türkiye has a small but significant contribution to chicken meat production.

In overall terms, Türkiye is above the world average in milk production and is an important producer. However, it has a smaller share in meat production. By diversifying and increasing the amount of meat production, Türkiye can gain a bigger place in the world market. Although it has a small share in chicken meat production compared to the world average, it still stands out as an important producer.

Although Türkiye has a good position in some areas of animal production, it has the potential for further development especially in meat production. According to these data, the animal production industry has strengths and needs to be improved. Especially increasing diversity and improving productivity in meat production can make Türkiye more competitive in the world market.

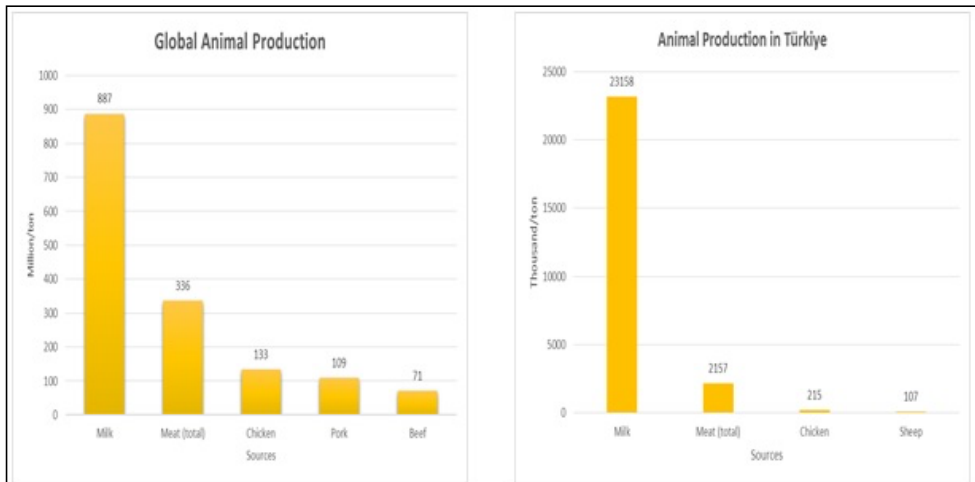


Figure 3: Animal production data by species in worldwide and Türkiye (18;19)

Animal Nutrition

Livestock, like all other living organisms, need nutrients in order to survive, repair their tissues, grow, reproduce, maintain their general welfare and produce. In order to provide these requirements, scientific feeding practices should be preferred. In nutrition, feed containing protein, carbohydrate, fat, water, vitamins and minerals are needed. It aims to understand the nutritional requirements of animals, the effects of feed additives, digestive physiology and the effects of ration on animal metabolism. The nutrients and quantities needed may differ according to the production (meat, milk or egg) purposes.

The nutrition of ruminant animals, which provide a significant portion of the food needed by humanity, covers topics such as nutrient absorption, ration requirements, the role of natural bioactive compounds and the integration of nutrition with digestive physiology. Nutrient requirements determine the amounts of protein, vitamins, minerals and other nutrients that animals need for healthy development and reproduction. Feed additives examines the effects of additives such as vitamins, minerals, amino acids and probiotics added to animal feeds on animal health and performance. Digestive physiology investigates the functioning of the digestive system of animals and how different nutrients are digested and absorbed. Metabolism and ration analyses the effects of different feeds on the metabolic processes of animals and how they affect overall health.



Figure 4: Free-grazing dairy cows

The nutrient needs of ruminants vary according to the direction of production. The survival allowance corresponds to the amount of energy required for the animal to sustain its life without any production. While age, breed, sex and live weight are taken into consideration in fattening production, ration is formed by considering parameters such as age, breed, lactation number and milk fat amount in milk production. Feed costs cover 70-80% of the costs in animal feeding. Animals should be fed at the energy level needed to protect their health and to provide income in terms of production. With the constantly renewed and developing technology, research and development opportunities have also increased. In order to meet the increasing demand for animal protein, animal nutritionists, zootechnicians and researchers are constantly turning to alternative products in this field. In general; alfalfa, corn silage, cottonseed meal, sunflower seed meal, grain feeds and other forage crops are preferred. With the prohibition of antibiotics in animal nutrition by the EU in 2006, animal immunity enhancing feedstuffs have also been orientated. In this context, the products produced by bees, which are known for their natural habitat, have been included in animal nutrition and researched.

Utilisation of Bee Products in Animal Production

Livestock, such as poultry and rabbits, as well as cattle, need to fulfil the global need for protein-rich foods. In many developed countries, interest in functional, healthy and organic foods is constantly increasing.

Research on the use of bee products as nutritional supplements in animal production has generally shown positive effects on the health and productivity of animals. Propolis dietary supplementation has been studied on weight gain, immune response, laying activity, egg quality and physiological parameters of broilers and laying hens with positive results (20). A study in which bee pollen (20 g/kg feed) was added to a broiler diet at high ambient temperature resulted in a significant ($p < 0.05$) 15.6% increase in average daily gain compared to the control group (21). Bee supplements are potential alternatives to in-feed antibiotics currently used in poultry feed (22). According to the results of another research, it was reported that the use of bee products such as propolis, bee pollen, royal jelly and bee venom in ration, drinking water or by injection has many benefits for poultry and rabbits (23). It was stated that the addition of a certain dose of propolis-based product to the ration of young bulls did not affect yield perfor-

mance, carcass characteristics, dry matter consumption and total digestibility of nutrients and efficiency of microbial synthesis (24). In a experiment conducted on sheep, it was observed that propolis supplementation at 100 and 200 mg/kg DM levels tended to improve digestion coefficients, rumen microbial fermentation, milk yield and composition reflected in increased weaning weight (25). In a previous study on whether propolis has an effect on ruminal fermentation and thus on methane gas reduction, it was concluded that phenolic acids, flavonoids and terpene compounds are the most common bioactive components found in different propolis species. It was reported that propolis affects rumen microbial fermentation through more potent mechanisms of action that alter rumen microbial fermentation and confer better antioxidant status and improved animal performance and health. It was emphasised that it positively affects rumen microbial fermentation and produces less methane formation (26).

Conclusion

This book chapter highlights the potential of using bee products in animal production and nutrition. It has been suggested that natural sources of bee products, especially propolis, bee pollen, royal jelly and bee venom, can be an effective alternative in animal nutrition. Research has shown that these products can have positive effects on animal health, immune system, weight gain, productivity and physiological parameters. For example, propolis supplementation has been shown to have positive results on factors such as weight gain, immune response and egg quality of broilers and laying hens.

In the view of these studies, the use of bee products in animal production may also be important in terms of sustainability and environmental impact. Since antibiotics are banned in animal nutrition, the importance of natural and organic alternatives has increased. In this respect, bee products can be a valuable resource to protect the health of animals and increase productivity. However, more research and application is necessary.

In Conclusion, the potential for the use of bee products in animal production is great and full utilisation of this potential is important both for the health and welfare of animals and for meeting the food security and nutritional needs of humans. Therefore, further research and application of bee products in the animal nutrition industry is required.

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EVALUATION OF THE RELATIONSHIP BETWEEN NUTRITION AND DISEASE IN HONEY BEES

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Introduction

Honey bees are vital pollinators essential for ecosystem balance and agricultural productivity. Their well-being is intricately linked to the health of our environment and food systems. Central to their vitality is the delicate balance between nutrition and disease susceptibility. Understanding the dynamic relationship between these factors is imperative for ensuring the resilience of honey bee populations and the ecosystems they support(1, 2).

Nutrition stands as a cornerstone of honey bee health, profoundly influencing their physiological resilience and immune response. A diverse and balanced diet, primarily sourced from floral nectar and pollen, provides bees with essential nutrients crucial for growth, development, and disease resistance. However, modern agricultural practices and environmental changes pose significant challenges to bee nutrition. Monoculture farming and habitat degradation can

restrict bee access to diverse forage, leading to nutrient deficiencies and compromised immune function. Consequently, weakened colonies are more susceptible to disease outbreaks, threatening both bee populations and the ecosystems they sustain(3).

Against this backdrop, comprehensively evaluating the nutritional landscape and disease dynamics of honey bee populations becomes paramount. Such assessments not only shed light on the vulnerabilities of bee populations but also offer insights into effective mitigation strategies. By promoting habitat diversity, implementing sustainable land management practices, and supporting bee-friendly agricultural initiatives, we can bolster bee nutrition and enhance their resilience against diseases(44, 43).

In summary, the intricate interplay between nutrition and disease underscores the critical importance of holistic approaches to honey bee health management. By addressing the root causes of nutritional imbalances and disease susceptibility, we can safeguard honey bee populations, preserve ecosystem integrity, and ensure food security for generations to come(4).

The health of honey bee populations is a matter of significant concern due to their vital role in pollination and ecosystem sustainability. Central to their well-being is the intricate interplay between nutrition and disease. As honey bees forage for nectar and pollen to sustain their colonies, they are exposed to various environmental stressors and pathogens, influencing their overall health and resilience(42, 5).

Nutrition serves as the foundation of honey bee vitality, fueling their metabolic processes and immune function. Nectar provides the energy needed for flight and daily activities, while pollen supplies essential proteins, lipids, vitamins, and minerals necessary for growth and development. However, the availability and diversity of floral resources are increasingly threatened by habitat loss, pesticide use, and climate change, compromising bee nutrition and immune function(42, 40).

Consequently, honey bee populations are vulnerable to a range of diseases and parasites. Pathogens such as *Nosema* and viruses, coupled with pests like the Varroa mite, pose significant threats to colony health. These stressors weaken bees, making them more susceptible to infections and colony collapse disorder, further exacerbating the decline in bee populations worldwide(41, 6).

Given the critical role of honey bees in ecosystem health and food security, understanding the complex relationship between nutrition and disease is paramount. By identifying key stressors and implementing sustainable management practices, we can mitigate the impact of nutritional deficiencies and disease outbreaks on honey bee populations. Through collaborative research efforts and informed conservation strategies, we can work towards safeguarding honey bee health and preserving the invaluable services they provide to ecosystems and agriculture(40, 7).

Honey bees are indispensable pollinators vital for maintaining ecosystem balance and agricultural productivity. However, their populations face numerous threats, with nutrition and disease being among the most significant factors affecting their health and survival. Understanding the intricate relationship between nutrition and disease in honey bee colonies is crucial for devising effective strategies to mitigate these threats and ensure their long-term well-being(39, 38).

Nutrition plays a fundamental role in honey bee health, influencing their physiological development, immune response, and overall resilience. Nectar and pollen obtained from flowering plants serve as the primary sources of nutrients for bees, providing essential carbohydrates, proteins, lipids, vitamins, and minerals. However, the availability and quality of these floral resources can vary significantly due to factors such as habitat loss, pesticide exposure, and changing environmental conditions, leading to nutritional deficiencies and compromised immune function in honey bee colonies(37, 8).

In addition to nutritional challenges, honey bees are also vulnerable to a wide range of diseases and parasites. Pathogens such as bacteria, viruses, fungi, and microsporidia, along with pests like the Varroa mite, pose serious threats to bee health and colony survival. These stressors can weaken bees, making them more susceptible to infections and contributing to population declines observed in recent years(36).

Honey bees play a critical role in the sustainability of ecosystems and agriculture, making them indispensable organisms for the environment. Their health is crucial not only for honey production but also for the pollination process of plants and overall biological diversity. Therefore, examining the nutrition status and disease relationships of honey bees is vital for both the beekeeping industry

and the preservation of ecosystems. The nutrition status of bees directly affects their immune systems and overall health. Inadequate or unbalanced nutrition can lead to diseases and weaken bee colonies(9, 10). Therefore, understanding the feeding habits and food sources of bees is a key step for beekeepers in maintaining bee health. Honey bees primarily feed on plant-based sources, such as nectar and pollen, which provide essential nutrients such as energy, protein, vitamins, and minerals. However, seasonal and geographical changes can affect these food sources and disrupt the nutritional balance of bees. Agricultural practices like monoculture can restrict bees' access to various nutrients and lead to nutritional deficiencies. These deficiencies weaken the immune systems of bee colonies, making them less resistant to diseases. For example, protein deficiency can adversely affect the development of bee larvae and weaken the overall strength of the colony(35, 34, 33). Additionally, imbalanced nutrition can decrease bees' resilience, leading to faster spread of diseases among colonies. In this context, evaluating the feeding habits and food sources of bees plays a vital role in preventing and controlling bee diseases. Adjusting nutrition and providing various food sources can help bees stay healthy and support the sustainability of the beekeeping industry. This introduction highlights the importance and complexity of the relationship between nutrition and disease, explaining why understanding this relationship is critical for maintaining bee health(11, 12).

Bees enhance agricultural productivity by pollinating many plants, but inadequate nutrition or diseases in bee colonies can lead to population declines and reduced pollination services. In recent years, the rapid decline in bee populations and the increase in bee diseases have been particularly concerning. This situation poses economic risks for the beekeeping industry, reducing agricultural productivity(13, 14, 15). Understanding and managing the relationship between bee nutrition and disease is necessary to protect bee populations and maintain agricultural productivity. Researchers, beekeepers, and environmental conservationists conduct various studies to better understand bee feeding habits, food sources, and disease risks. These studies can contribute to the development of effective beekeeping management strategies and the preservation of ecosystem health. Preserving bee health is critical not only for the sustainability of the beekeeping industry but also for maintaining the balance of natural ecosystems(32). Therefore, understanding and managing the relationship between nutrition and disease are necessary. Evaluating the relationship between bee nu-

trition and disease is also important for human health. Bees contribute to food security and diversity by aiding in the reproduction of many plants. Therefore, managing bee feeding habits and disease risks is crucial for food security and human health. Factors such as pesticide use, habitat loss, and climate change can adversely affect bee feeding habits and immune systems. This can reduce bees' resistance to diseases, leading to colony losses and instability in the beekeeping industry. Understanding and managing the relationship between nutrition and disease can help protect bee health and populations by reducing these risk factors(16). Therefore, evaluating and managing bee nutrition habits and disease risks is of great importance to beekeepers and communities. Research and practical applications in this field play a critical role in preserving bee health and ecosystem balance. This introduction emphasizes the critical importance of understanding the relationship between bee nutrition and disease for human health, food security, and environmental sustainability. Bee health is a fundamental factor for both ecosystems and human well-being, so managing the relationship between nutrition and disease is crucial(31, 30). Evaluating the relationship between bee nutrition and disease not only helps understand the current situation but also prepares for future threats. Environmental factors such as climate change and the emergence of new pathogens and disease types should also be considered. Continuous research and knowledge accumulation are vital for beekeepers and the beekeeping industry. Additionally, collaboration and knowledge sharing between beekeepers and scientists are necessary. While beekeepers can support scientific studies with field observations and experiences, scientists can develop solutions tailored to the needs of beekeepers(17, 18). This collaboration facilitates a better understanding of the relationship between bee nutrition and disease and the development of effective management strategies. In conclusion, evaluating the relationship between bee nutrition and disease is a complex issue that requires the participation of multiple stakeholders. However, with the right approaches and strategies, bee health can be preserved, and the sustainability of the beekeeping industry ensured. Continuous research and experience sharing are important for better understanding and managing this critical issue. Assessing the relationship between bee nutrition and disease is essential not only for bees and beekeepers but also for the health of ecosystems and biological diversity(19, 20). Bees play a crucial role in facilitating the reproduction of many plants in various ecosystems. Therefore, declines in bee

populations or disease outbreaks can disrupt the balance of natural life and negatively impact ecosystem functionality. The health of bee populations in agricultural areas is also important for the productivity and diversity of agricultural products. Bee pollination services support the production of fruits, vegetables, and other plant products. Therefore, the nutrition status and disease resistance of bees have a direct impact on agricultural production. Preserving bee health is critical for the sustainability and diversity of food production. This introduction section outlines the basic concepts and importance required to evaluate the relationship between honey bee nutrition and disease(21, 22). Maintaining bee health and protecting them from diseases are essential for the sustainability of ecosystems, agricultural production, and overall biological diversity. Therefore, understanding and managing bee feeding habits and disease risks are necessary. Studies in this field will contribute to the preservation of bee health and the sustainability of the beekeeping industry(29).

1. Effects of Nutrition on Bees

Health Nutrition affects the health of bees both at the individual and colony levels. Among the primary food sources are pollen and nectar. Pollen is rich in protein, fat, vitamins, and minerals, crucial for the growth, development, and reproduction functions of bees. Nectar primarily serves as a carbohydrate source, meeting bees' energy needs(28).

1.1. Importance of Pollen

Pollen is the main source of protein and other nutrients critical for bees' survival and colony health. Inadequate pollen intake can adversely affect larval development and weaken the immune system of adult bees, creating a colony structure vulnerable to diseases(23).

1.2. Nectar and Energy Requirement

Nectar provides the necessary energy for bees' daily activities and colony survival during winter months. Inadequate energy intake can limit bees' foraging behavior and their ability to maintain colony internal temperature in cold weather conditions. This can lead to colony weakness or even collapse(27).

1.3. Nutritional Diversity and Disease Resistance

Nutritional diversity significantly affects bees' overall health. Access to various pollen and nectar sources allows bees to have a more balanced diet and benefit from a wide range of nutrients. This can help bees become more resistant to diseases(26, 25).

2. Combatting Diseases and Pests

Bees are confronted with both internal and external pathogens. These pathogens include bacteria, viruses, fungi, and parasites. Particularly, the Varroa destructor mite poses a significant threat to bee colonies worldwide(44, 24).

2.1 Effects of Varroa Destructor

The Varroa mite feeds on the hemolymph (blood-like fluid) of bees, transmitting various viruses. This weakens infected bees and can lead to colony collapse. Controlling Varroa is done through chemical acaricides, biological methods, and colony management practices(15).

2.2. Other Diseases and Pests

Among other significant diseases and pests are American foulbrood, European foulbrood, and microsporidia such as Nosema sp. Effective management of these pathogens is generally achieved through medication treatment, colony hygiene, and regular colony inspections. The relationship between bee feeding and diseases is a fundamental issue for their health and environmental interactions. Promoting healthy feeding habits and effectively combating diseases are essential for the conservation of bee populations and sustainable agricultural production. Research in this field will contribute to a better understanding and preservation of bee health, thus contributing to the maintenance of ecological balance and food security(17).

3. Feeding Strategies and Management Approaches

Adopting correct feeding strategies is critical for maintaining the health and productivity of bee colonies. Colony managers and beekeepers implement

various methods to provide bees with the necessary nutrients adequately and in a balanced manner(14).

3.1. Artificial Feeding

During periods of insufficient natural resources or when combating specific diseases, artificial food supplements can be provided to bees. These supplements usually come in the form of sugar syrup or pollen substitutes, meeting the energy and protein needs of the bees. However, artificial feeding must be done correctly; excessive or improper artificial feeding can adversely affect bees' ability to forage for natural food sources.

3.2. Diversity and Habitat Management

Preserving and promoting the diversity of flowering plants is essential to increase the natural feeding diversity of bees. Beekeepers and environmental managers can work to plant bee-friendly species and encourage a variety of long-flowering plant species. This enhances both the quality of bee nutrition and supports ecological diversity(21).

3.3. Regulation of Pesticide and Chemical Use

The use of pesticides and chemicals on bee colonies can directly affect bee health. The use of pesticides, fungicides, and herbicides can leave residues on nectar and pollen, preventing bees from benefiting from these food sources. The use of such chemicals should be subject to specific regulations, and whenever possible, bee-friendly alternatives should be preferred(28).

4. Monitoring and Evaluation Methods for Bee Health

Regular monitoring and evaluation methods should be implemented to assess the effectiveness of interventions in bee health and to detect disease symptoms at an early stage(2).

4.1. Colony Inspections

Regular colony inspections play a fundamental role in understanding bee health and colony dynamics. During these inspections, the development of larvae, the

condition of adult bees, and the overall colony activity are observed. Important parameters such as disease symptoms, parasite infestations, and feeding status are determined through these inspections(5, 12).

4.2. Early Intervention against Diseases and Pests

Early diagnosis and intervention are crucial to prevent the spread of diseases and pests and to maintain colony health. Particularly for pests like Varroa destructor, regular monitoring and appropriate treatment methods are of great importance(9).

5. Conclusion and Recommendations

The relationship between bee feeding and diseases serves as an important indicator for the sustainability of ecosystems and agricultural production. Comprehensive understanding and multifaceted strategies are required to support bee health and effectively combat diseases. Beekeepers, environmental managers, and policymakers can contribute to this process by adopting and implementing supportive policies for bees. Taking steps to preserve bee populations and ensure natural diversity and food security can leave a healthier and more balanced environment for future generations(28, 31).

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